

UNIVERSITÄT BONN

Harmonic Analysis Notes

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Part I

Foundations of Harmonic Analysis

1 Interpolation Theorems

The notes proceed from interpolation to Fourier summation and approximate identities, then to maximal functions and differentiation, and finally to ergodic theory. Throughout, we use the notation

$$e(x) = e^{2\pi i x}, \quad \mathbb{T} = \mathbb{R}/\mathbb{Z}, \quad \widehat{f}(n) = \int_0^1 f(x) e^{-2\pi i n x} dx,$$

and $|E|$ denotes Lebesgue measure. For a measure-preserving transformation T , we write

$$A_m f = \frac{1}{m} \sum_{k=0}^{m-1} f \circ T^k.$$

1.1 The Riesz–Thorin Interpolation Theorem

Theorem 1.1 (Riesz–Thorin). *Let $1 \leq p_1, p_2, q_1, q_2 \leq \infty$, let (X, μ) and $(\tilde{X}, \tilde{\mu})$ be measure spaces, and let T be a linear operator such that*

$$T : L^{p_1}(X) \longrightarrow L^{q_1}(\tilde{X}), \quad T : L^{p_2}(X) \longrightarrow L^{q_2}(\tilde{X})$$

is bounded with norms M_1 and M_2 , respectively. If

$$\left(\frac{1}{p}, \frac{1}{q}\right) = (1-\theta) \left(\frac{1}{p_1}, \frac{1}{q_1}\right) + \theta \left(\frac{1}{p_2}, \frac{1}{q_2}\right), \quad 0 \leq \theta \leq 1,$$

then $T : L^p(X) \rightarrow L^q(\tilde{X})$ is bounded and

$$\|T\|_{L^p \rightarrow L^q} \leq M_1^{1-\theta} M_2^\theta.$$

Proof sketch via the three-lines lemma. For simple functions f and g , consider

$$\Phi(z) = \int_{\tilde{X}} T(f_z) g_z d\tilde{\mu},$$

where

$$f_z = |f|^{p \left(\frac{1-z}{p_1} + \frac{z}{p_2} \right)} \frac{f}{|f|}, \quad g_z = |g|^{q' \left(\frac{1-z}{q'_1} + \frac{z}{q'_2} \right)} \frac{g}{|g|}.$$

The function Φ is holomorphic in the strip $0 < \operatorname{Re} z < 1$, bounded and continuous on its closure, and the endpoint assumptions give

$$|\Phi(iy)| \leq M_1, \quad |\Phi(1+iy)| \leq M_2.$$

Hadamard's three-lines lemma yields

$$|\Phi(\theta)| \leq M_1^{1-\theta} M_2^\theta.$$

Taking the supremum over g and using duality proves the claim. $\square$

Remark 1.2. Linearity is essential in the Riesz–Thorin theorem. The interpolated indices lie on

the line segment joining $(1/p_1, 1/q_1)$ and $(1/p_2, 1/q_2)$.

1.2 Sublinear Operators and Weak L^p Spaces

Definition 1.3. An operator T , defined on a dense subclass $\mathcal{D} \subseteq L^p(X)$, is *sublinear* if

$$|T(f_1 + f_2)| \leq |Tf_1| + |Tf_2|, \quad |T(cf)| = |c| |Tf|.$$

Definition 1.4. For $1 \leq p < \infty$, the weak L^p quasi-norm is

$$\|f\|_{L^{p,\infty}(X,\mu)} = \sup_{\lambda>0} \lambda \mu(\{x \in X : |f(x)| > \lambda\})^{1/p}.$$

We write $L^{\infty,\infty} = L^\infty$.

This is only a quasi-norm: its triangle inequality holds up to a constant. Chebyshev's inequality gives the continuous embedding $L^p \subseteq L^{p,\infty}$:

$$\mu\{|f| > \lambda\} \leq \int_{\{|f|>\lambda\}} \frac{|f|^p}{\lambda^p} d\mu \leq \frac{\|f\|_{L^p}^p}{\lambda^p},$$

and hence

$$\|f\|_{L^{p,\infty}} \leq \|f\|_{L^p}.$$

The converse is false. For example,

$$|x|^{-d/p} \in L^{p,\infty}(\mathbb{R}^d) \setminus L^p(\mathbb{R}^d).$$

The notation $T : L^p(X) \rightarrow L^{q,\infty}(\tilde{X})$ means that

$$\|Tf\|_{L^{q,\infty}} \leq C \|f\|_{L^p};$$

this is called a weak-type estimate.

1.3 The Marcinkiewicz Interpolation Theorem

Theorem 1.5 (Marcinkiewicz). *Let T be sublinear, let $1 \leq p_1, p_2, q_1, q_2 \leq \infty$, assume $p_j \leq q_j$ and $q_1 \neq q_2$, and suppose*

$$T : L^{p_1} \longrightarrow L^{q_1,\infty}, \quad T : L^{p_2} \longrightarrow L^{q_2,\infty}$$

are weak-type bounds. If

$$\left(\frac{1}{p}, \frac{1}{q}\right) = (1-\theta) \left(\frac{1}{p_1}, \frac{1}{q_1}\right) + \theta \left(\frac{1}{p_2}, \frac{1}{q_2}\right), \quad 0 < \theta < 1,$$

then $T : L^p \rightarrow L^q$ is strongly bounded.

Remark 1.6. In contrast with Riesz–Thorin, Marcinkiewicz interpolation applies to sublinear operators and assumes only weak-type endpoint bounds. The price is that the endpoints are excluded and the interpolation segment cannot be horizontal.

We give the proof in the most frequently used diagonal case $p_j = q_j$. Suppose

$$p_1 < p < p_2 < \infty, \quad \|Tf\|_{L^{p_j,\infty}} \leq A_j \|f\|_{p_j}.$$

For every measurable g , the layer-cake formula gives

$$\|g\|_{L^p}^p = p \int_0^\infty \lambda^{p-1} \mu\{|g| > \lambda\} d\lambda.$$

For a fixed $\lambda > 0$, split f at height λ :

$$f_1^\lambda = f \mathbf{1}_{\{|f| > \lambda\}}, \quad f_2^\lambda = f \mathbf{1}_{\{|f| \leq \lambda\}}.$$

On $\{|f| > \lambda\}$,

$$|f|^{p_1} = |f|^p |f|^{p_1-p} \leq |f|^p \lambda^{p_1-p},$$

so $f_1^\lambda \in L^{p_1}$. Similarly,

$$|f|^{p_2} \leq |f|^p \lambda^{p_2-p} \quad \text{on } \{|f| \leq \lambda\},$$

and therefore $f_2^\lambda \in L^{p_2}$.

Sublinearity and the weak-type estimates imply

$$\begin{aligned} \mu\{|Tf| > 2\lambda\} &\leq \mu\{|Tf_1^\lambda| > \lambda\} + \mu\{|Tf_2^\lambda| > \lambda\} \\ &\leq \frac{A_1^{p_1} \|f_1^\lambda\|_{p_1}^{p_1}}{\lambda^{p_1}} + \frac{A_2^{p_2} \|f_2^\lambda\|_{p_2}^{p_2}}{\lambda^{p_2}}. \end{aligned}$$

Consequently, by Fubini's theorem,

$$\begin{aligned} \|Tf\|_p^p &\leq 2^p p \left[A_1^{p_1} \int_0^\infty \lambda^{p-p_1-1} \int_{\{|f| > \lambda\}} |f|^{p_1} d\mu d\lambda \right. \\ &\quad \left. + A_2^{p_2} \int_0^\infty \lambda^{p-p_2-1} \int_{\{|f| \leq \lambda\}} |f|^{p_2} d\mu d\lambda \right]. \end{aligned}$$

Changing the order of integration gives

$$\begin{aligned} \int_0^\infty \lambda^{p-p_1-1} \int_{\{|f| > \lambda\}} |f|^{p_1} d\mu d\lambda &= \frac{\|f\|_p^p}{p-p_1}, \\ \int_0^\infty \lambda^{p-p_2-1} \int_{\{|f| \leq \lambda\}} |f|^{p_2} d\mu d\lambda &= \frac{\|f\|_p^p}{p_2-p}. \end{aligned}$$

Thus

$$\|Tf\|_p \leq 2 \left[p \left(\frac{A_1^{p_1}}{p-p_1} + \frac{A_2^{p_2}}{p_2-p} \right) \right]^{1/p} \|f\|_p.$$

The constant blows up as $p \downarrow p_1$ or $p \uparrow p_2$, which explains why the endpoints are excluded.

If $p_2 = \infty$, split instead at height $\lambda/(2A_2)$. Then

$$\left\| T f_2^{\lambda/(2A_2)} \right\|_\infty \leq A_2 \frac{\lambda}{2A_2} = \frac{\lambda}{2},$$

so the second distribution-function term disappears, and the rest of the argument is unchanged.

1.4 A Standard Application Pattern

Maximal operators often fail to map L^1 to L^1 . The standard route to strong L^p bounds is:

1. prove $T : L^\infty \rightarrow L^\infty$;

2. prove the weak-type bound $T : L^1 \rightarrow L^{1,\infty}$, usually by a covering lemma;
3. interpolate to obtain strong type (p, p) for $1 < p < \infty$.

2 Fourier Series, Summability Kernels, and Approximate Identities

2.1 Convolution and the Physical–Frequency Dictionary

If $f, g \in L^1(\mathbb{T})$, then

$$\widehat{f * g}(n) = \widehat{f}(n)\widehat{g}(n).$$

The following correspondences are fundamental:

	Physical side	Frequency side
1	$f * g$	$\widehat{f}\widehat{g}$
2	f'	multiplication by $2\pi in$
3	multiplication by $1 - e^{2\pi ix}$	$\Delta \widehat{f}(n) = \widehat{f}(n) - \widehat{f}(n-1)$
4	regularity	decay
5	decay	regularity

For example, if $f \in C^k(\mathbb{T})$, repeated integration by parts and periodicity give

$$\widehat{f}(n) = \frac{\widehat{f^{(k)}}(n)}{(2\pi in)^k}, \quad |\widehat{f}(n)| \leq \frac{\|f^{(k)}\|_1}{(2\pi|n|)^k}.$$

Thus smoothness forces polynomial decay of Fourier coefficients. Conversely, suitable decay or summability assumptions on Fourier coefficients imply regularity. Multiplication by $1 - e^{2\pi ix}$, for example, corresponds to discrete differentiation of the Fourier coefficients.

2.2 The Dirichlet and Fejér Kernels

The N -th symmetric Fourier partial sum is

$$S_N f = f * D_N, \quad D_N(x) = \sum_{n=-N}^N e(n x).$$

On the frequency side, S_N is multiplication by $\mathbf{1}_{[-N, N]}$, a sharp cutoff. Accordingly, D_N oscillates, changes sign, and satisfies

$$\|D_N\|_{L^1(\mathbb{T})} \sim \log N.$$

This unboundedness is the source of the failure of $S_N f \rightarrow f$ in general in $C(\mathbb{T})$ and $L^1(\mathbb{T})$.

Cesàro averaging replaces the sharp cutoff by a smoother one:

$$\sigma_N f = \frac{1}{N} \sum_{k=0}^{N-1} S_k f = f * K_N, \quad K_N = \frac{1}{N} \sum_{k=0}^{N-1} D_k.$$

To compute the closed form, set $z = e(x)$. A geometric sum gives

$$D_k = \frac{z^{k+\frac{1}{2}} - z^{-k-\frac{1}{2}}}{z^{\frac{1}{2}} - z^{-\frac{1}{2}}}.$$

Hence

$$\begin{aligned}
K_N(z^{\frac{1}{2}} - z^{-\frac{1}{2}}) &= \frac{1}{N} \sum_{k=0}^{N-1} (z^{k+\frac{1}{2}} - z^{-k-\frac{1}{2}}) \\
&= \frac{1}{N} \left(\frac{z^N - 1}{z^{\frac{1}{2}} - z^{-\frac{1}{2}}} - \frac{1 - z^{-N}}{z^{\frac{1}{2}} - z^{-\frac{1}{2}}} \right) \\
&= \frac{1}{N} \frac{(z^{N/2} - z^{-N/2})^2}{z^{\frac{1}{2}} - z^{-\frac{1}{2}}}.
\end{aligned}$$

Here we used

$$\sum_{k=0}^{N-1} z^{k+\frac{1}{2}} = \frac{z^{\frac{1}{2}}(z^N - 1)}{z - 1} = \frac{z^N - 1}{z^{\frac{1}{2}} - z^{-\frac{1}{2}}}.$$

Therefore

$$K_N(x) = \frac{1}{N} \left(\frac{z^{N/2} - z^{-N/2}}{z^{1/2} - z^{-1/2}} \right)^2 = \frac{1}{N} \frac{\sin^2(\pi Nx)}{\sin^2(\pi x)}.$$

For $|x| \leq 1/2$,

$$0 \leq K_N(x) \lesssim \min \left\{ \frac{1}{Nx^2}, N \right\} = \frac{1}{N} \min \left\{ \frac{1}{x^2}, N^2 \right\}.$$

Indeed,

$$\sin^2(\pi Nx) \leq \min\{1, (\pi Nx)^2\}, \quad \sin^2(\pi x) \geq 4x^2$$

in this range.

On the frequency side, the Fejér kernel has a triangular cutoff:

$$K_N(x) = \sum_{|k| < N} \left(1 - \frac{|k|}{N} \right) e(kx).$$

This follows by counting:

$$\begin{aligned}
\frac{1}{N} \sum_{k=0}^{N-1} \sum_{|j| \leq k} e(jx) &= \frac{1}{N} \sum_{|j| \leq N-1} \#\{k : |j| \leq k \leq N-1\} e(jx) \\
&= \sum_{|j| < N} \left(1 - \frac{|j|}{N} \right) e(jx).
\end{aligned}$$

2.3 Approximate Identities

Definition 2.1. A sequence $\{\Phi_N\}_{N=1}^{\infty} \subseteq L^{\infty}(\mathbb{T})$ is an *approximate identity* if

1. $\int_0^1 \Phi_N(x) dx = 1$;
2. $\sup_N \int_0^1 |\Phi_N(x)| dx < \infty$;
3. for every $\delta > 0$,

$$\lim_{N \rightarrow \infty} \int_{|x| > \delta} |\Phi_N(x)| dx = 0.$$

Example 2.2. The Fejér kernels form an approximate identity. Their zeroth Fourier coefficient

gives $\int K_N = 1$. Since $K_N \geq 0$, $\int |K_N| = 1$. Finally,

$$\int_{|x|>\delta} |K_N(x)| dx \leq \frac{C}{N} \int_{\delta}^{1/2} \frac{dx}{x^2} \leq \frac{C}{N\delta} \rightarrow 0.$$

By contrast, the Dirichlet kernels violate the second condition because $\|D_N\|_1 \sim \log N$.

2.4 Convergence of Approximate Identities

Proposition 2.3. *Let $\{\Phi_N\}$ be an approximate identity.*

1. *If $f \in C(\mathbb{T})$, then $\|\Phi_N * f - f\|_{L^\infty} \rightarrow 0$.*
2. *If $f \in L^p(\mathbb{T})$, $1 \leq p < \infty$, then $\|\Phi_N * f - f\|_{L^p} \rightarrow 0$.*
3. *If $\mu \in M(\mathbb{T})$, then $\mu * \Phi_N \rightarrow \mu$ in the weak-* topology.*

Proof. For the first assertion, normalization gives

$$\Phi_N * f(x) - f(x) = \int_{\mathbb{T}} \Phi_N(y) (f(x-y) - f(x)) dy.$$

Split this integral into $|y| \leq \delta$ and $|y| > \delta$. Uniform continuity of f gives, for sufficiently small δ ,

$$\int_{|y| \leq \delta} |\Phi_N(y)| |f(x-y) - f(x)| dy \leq \varepsilon \|\Phi_N\|_1 \leq C\varepsilon.$$

The remaining part is bounded by

$$2 \|f\|_\infty \int_{|y|>\delta} |\Phi_N(y)| dy,$$

which tends to zero.

For $1 \leq p < \infty$, Minkowski's integral inequality gives

$$\|\Phi_N * f - f\|_{L^p} \leq \int_{\mathbb{T}} |\Phi_N(y)| \|f(\cdot - y) - f\|_{L^p} dy.$$

Translations are continuous in L^p :

$$\|f(\cdot - y) - f\|_{L^p} \rightarrow 0 \quad \text{as } y \rightarrow 0.$$

Splitting again at δ proves the second assertion.

For the third assertion, define $\tilde{\Phi}_N(y) = \Phi_N(-y)$. Fubini's theorem gives, for $f \in C(\mathbb{T})$,

$$\int_{\mathbb{T}} f d(\mu * \Phi_N) = \int_{\mathbb{T}} (\tilde{\Phi}_N * f) d\mu \rightarrow \int_{\mathbb{T}} f d\mu,$$

using the uniform convergence from the first assertion and the finiteness of $|\mu|(\mathbb{T})$. $\square$

Corollary 2.4 (Fejér's theorem). *If $f \in C(\mathbb{T})$, then $\sigma_N f \rightarrow f$ uniformly. If $f \in L^p(\mathbb{T})$, $1 \leq p < \infty$, then $\sigma_N f \rightarrow f$ in L^p . In particular, trigonometric polynomials are dense in both $C(\mathbb{T})$ and $L^p(\mathbb{T})$.*

3 The Hardy–Littlewood Maximal Function

3.1 Motivation: A Generalized Newton–Leibniz Formula

Let $f : [a, b] \rightarrow \mathbb{R}$ be integrable and define

$$F(x) = \int_a^x f(y) dy.$$

Formally,

$$F'(x) = \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{|I|} \int_I f, \quad I = (x, x+h).$$

This raises the question whether shrinking averages converge almost everywhere to $f(x)$. In $\mathbb{R}^d$, the corresponding question is whether

$$\lim_{\substack{|B| \rightarrow 0 \\ x \in B}} \frac{1}{|B|} \int_B f(y) dy = f(x) \quad \text{for almost every } x.$$

The standard tool for controlling such families of averages is the maximal function.

3.2 Definition and Main Estimate

Definition 3.1. For $f \in L^1(\mathbb{R}^d)$, the uncentered Hardy–Littlewood maximal function is

$$Mf(x) = \sup_{B \ni x} \frac{1}{|B|} \int_B |f(y)| dy,$$

where the supremum is over all open balls containing x .

Theorem 3.2. If $f \in L^1(\mathbb{R}^d)$, then:

- (i) Mf is measurable;
- (ii) $Mf(x) < \infty$ for almost every x ;
- (iii) for every $\alpha > 0$,

$$|\{x : Mf(x) > \alpha\}| \leq \frac{3^d}{\alpha} \|f\|_{L^1}.$$

The last inequality is the weak-type $(1, 1)$ estimate. In general $Mf \notin L^1$ unless $f = 0$, so weak type is the best possible endpoint result.

3.3 Proof of the Weak-Type Estimate

For $\alpha > 0$, set

$$E_\alpha = \{x \in \mathbb{R}^d : Mf(x) > \alpha\}.$$

If $x \in E_\alpha$, there is an open ball $B \ni x$ such that

$$\frac{1}{|B|} \int_B |f| > \alpha.$$

The same ball works for every $x' \in B$, so $B \subseteq E_\alpha$. Thus E_α is open, Mf is lower semicontinuous, and in particular it is measurable.

Lemma 3.3 (Finite Vitali-type covering lemma). *Let $\mathcal{B} = \{B_1, \dots, B_n\}$ be a finite family of open balls. There is a pairwise disjoint subfamily $B_{i_1}, \dots, B_{i_k}$ such that*

$$\left| \bigcup_{i=1}^n B_i \right| \leq 3^d \sum_{p=1}^k |B_{i_p}|.$$

Proof. Choose a ball of largest radius and discard all balls intersecting it. Repeat the procedure with the remaining balls. The selected balls are pairwise disjoint. Every discarded ball B_j intersects a selected ball B_{i_p} whose radius is at least r_j , and hence

$$B_j \subseteq 3B_{i_p}.$$

Therefore

$$\left| \bigcup_i B_i \right| \leq \left| \bigcup_p 3B_{i_p} \right| \leq \sum_p |3B_{i_p}| = 3^d \sum_p |B_{i_p}|.$$

□

For each $x \in E_\alpha$, choose a ball $B_x \ni x$ satisfying

$$|B_x| < \frac{1}{\alpha} \int_{B_x} |f|.$$

Let $K \subseteq E_\alpha$ be compact. By compactness, finitely many of the B_x cover K . Apply the covering lemma and retain a pairwise disjoint subfamily $\{B_{i_p}\}$. Then

$$|K| \leq 3^d \sum_p |B_{i_p}| \leq \frac{3^d}{\alpha} \sum_p \int_{B_{i_p}} |f| \leq \frac{3^d}{\alpha} \|f\|_{L^1}.$$

Inner regularity of Lebesgue measure now yields

$$|E_\alpha| \leq \frac{3^d}{\alpha} \|f\|_{L^1}.$$

Finally,

$$|\{Mf = \infty\}| \leq |\{Mf > \alpha\}| \leq \frac{3^d}{\alpha} \|f\|_1 \rightarrow 0$$

as $\alpha \rightarrow \infty$, proving almost-everywhere finiteness.

Corollary 3.4. *For $1 < p \leq \infty$,*

$$\|Mf\|_{L^p} \leq C_{d,p} \|f\|_{L^p}.$$

Proof. The L^∞ estimate is immediate, and the weak $(1, 1)$ estimate was proved above. Apply Marcinkiewicz interpolation. □

3.4 The Lebesgue Differentiation Theorem

Theorem 3.5 (Lebesgue differentiation theorem). *For $f \in L^1(\mathbb{R}^d)$,*

$$\lim_{\substack{|B| \rightarrow 0 \\ x \in B}} \frac{1}{|B|} \int_B f(y) dy = f(x) \quad \text{for almost every } x.$$

Proof. For $\alpha > 0$, let

$$E_\alpha = \left\{ x : \limsup_{\substack{|B| \rightarrow 0 \\ x \in B}} \left| \frac{1}{|B|} \int_B f - f(x) \right| > \alpha \right\}.$$

It suffices to prove $|E_\alpha| = 0$.

Choose $g \in C_c^\infty(\mathbb{R}^d)$ such that $\|g - f\|_1 < \varepsilon$. By the triangle inequality,

$$\begin{aligned} \left| \frac{1}{|B|} \int_B f - f(x) \right| &\leq \frac{1}{|B|} \int_B |f - g| + \left| \frac{1}{|B|} \int_B g - g(x) \right| + |g(x) - f(x)| \\ &\leq (f - g)^*(x) + \left| \frac{1}{|B|} \int_B g - g(x) \right| + |g(x) - f(x)|. \end{aligned}$$

As $|B| \rightarrow 0$, the middle term tends to zero by continuity of g . Therefore

$$E_\alpha \subseteq \left\{ (f - g)^* > \frac{\alpha}{2} \right\} \cup \left\{ |g - f| > \frac{\alpha}{2} \right\}.$$

The weak $(1, 1)$ estimate and Chebyshev's inequality give

$$|E_\alpha| \leq \frac{2 \cdot 3^d}{\alpha} \|f - g\|_1 + \frac{2}{\alpha} \|f - g\|_1 \leq \frac{2(3^d + 1)}{\alpha} \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, $|E_\alpha| = 0$. □

3.5 Lebesgue Points

The differentiation theorem implies

$$Mf(x) \geq |f(x)| \quad \text{for almost every } x.$$

It also extends to $f \in L_{\text{loc}}^1(\mathbb{R}^d)$ by localization: on $B(0, R)$, replace f by $f \mathbf{1}_{B(0, 2R)}$.

Definition 3.6. A point x is a *Lebesgue point* of f if

$$\lim_{\substack{|B| \rightarrow 0 \\ x \in B}} \frac{1}{|B|} \int_B |f(y) - f(x)| dy = 0.$$

Corollary 3.7. If $f \in L_{\text{loc}}^1(\mathbb{R}^d)$, then almost every point of $\mathbb{R}^d$ is a Lebesgue point of f .

Proof. For each $q \in \mathbb{Q}$, apply the differentiation theorem locally to $|f - q|$, and let N_q be the corresponding null exceptional set. The union $N = \bigcup_{q \in \mathbb{Q}} N_q$ is still null. If $x \notin N$ and $\varepsilon > 0$, choose $q \in \mathbb{Q}$ such that $|f(x) - q| < \varepsilon$. Then

$$\begin{aligned} \limsup_{\substack{|B| \rightarrow 0 \\ x \in B}} \frac{1}{|B|} \int_B |f - f(x)| &\leq \limsup_{\substack{|B| \rightarrow 0 \\ x \in B}} \frac{1}{|B|} \int_B |f - q| + \varepsilon \\ &= |f(x) - q| + \varepsilon < 2\varepsilon. \end{aligned}$$

Letting $\varepsilon \rightarrow 0$ proves the result. □

4 Ergodic Theory

4.1 Measure-Preserving Dynamical Systems

Let $(X, \mathcal{A}, \mu)$ be a probability space. Suppose $T : X \rightarrow X$ is measurable and invertible, T^{-1} is measurable, and T preserves measure:

$$\mu(T^{-1}A) = \mu(A) \quad \text{for all } A \in \mathcal{A}, \quad \text{equivalently} \quad T_*\mu = \mu.$$

Then (X, T) is a measure-preserving $\mathbb{Z}$ -system. Its two-sided orbits are

$$\dots, T^{-2}x, T^{-1}x, x, Tx, T^2x, \dots$$

The associated Koopman operator is

$$U_T : L^2(X) \longrightarrow L^2(X), \quad U_T f = f \circ T.$$

Measure preservation implies

$$\int_X f \circ T \, d\mu = \int_X f \, d\mu$$

and hence

$$\int_X |f \circ T|^2 \, d\mu = \int_X |f|^2 \, d\mu.$$

Thus U_T is an isometry; if T is invertible, U_T is unitary.

Example 4.1. For a torus rotation, take

$$\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d, \quad T(x) = x + \alpha \pmod{\mathbb{Z}^d}.$$

This is a model of structured dynamics.

Example 4.2. For a Bernoulli system, take

$$X = \{0, 1\}^{\mathbb{Z}}$$

with a product probability measure and let T be the shift. If $T((x_n)_{n \in \mathbb{Z}}) = (y_n)_{n \in \mathbb{Z}}$, then $y_{n+1} = x_n$. This is a model of random dynamics.

4.2 Ergodicity

Definition 4.3. The system (X, T) is *ergodic* if every measurable invariant set is trivial:

$$TA = A \text{ a.e.} \implies \mu(A) \in \{0, 1\}.$$

Equivalently, every invariant L^2 function is constant:

$$U_T f = f \implies f \text{ is constant almost everywhere.}$$

Indeed, if the set formulation holds and $U_T f = f$, every superlevel set $\{f > c\}$ is invariant and has measure zero or one. This forces f to be constant. Conversely, apply the functional statement to $\mathbf{1}_A$.

4.3 The von Neumann Mean Ergodic Theorem

Theorem 4.4 (von Neumann mean ergodic theorem). *Let (X, T) be an ergodic measure-preserving probability system and let $f \in L^2(X)$. Then*

$$\underbrace{\frac{1}{N} \sum_{n=0}^{N-1} f \circ T^n}_{\text{time average}} \xrightarrow{L^2} \underbrace{\mathbb{E}[f] \mathbf{1}}_{\text{space average}}.$$

Proof. Write

$$A_N f = \frac{1}{N} \sum_{n=0}^{N-1} U_T^n f.$$

Since U_T is an isometry, $\|A_N\|_{L^2 \rightarrow L^2} \leq 1$. Let

$$C = \overline{\text{conv}}\{f \circ T^n : n \geq 0\} \subseteq L^2.$$

This is a nonempty closed convex set. By the nearest-point property in Hilbert spaces, C contains a unique element g of minimal norm.

The operator U_T maps the orbit into itself and therefore maps C into C . Since it is an isometry, $U_T g$ has the same minimal norm as g . Uniqueness gives $U_T g = g$, and ergodicity implies that $g \equiv c$ is constant.

The functional $h \mapsto \int_X h d\mu$ is continuous on L^2 because $\mu(X) = 1$. It has the value $\int f d\mu$ on every element of the orbit, hence on all of C . In particular,

$$c = \mathbb{E}[c] = \mathbb{E}[f].$$

It remains to show that $A_N f \rightarrow g$. The key estimate is

$$\|A_N(U_T f) - A_N f\|_2 = \frac{1}{N} \|f \circ T^N - f\|_2 \leq \frac{2}{N} \|f\|_2.$$

For a finite convex combination

$$h = \sum_{n=0}^M a_n U_T^n f, \quad a_n \geq 0, \quad \sum_{n=0}^M a_n = 1,$$

iteration gives

$$\|A_N h - A_N f\|_2 \leq \frac{2M \|f\|_2}{N}.$$

Given $\varepsilon > 0$, choose such an h with $\|h - g\|_2 < \varepsilon$. Because $A_N g = g$,

$$\|A_N f - g\|_2 \leq \|A_N f - A_N h\|_2 + \|A_N(h - g)\|_2 \leq \frac{2M \|f\|_2}{N} + \varepsilon.$$

Taking the limit superior and then letting $\varepsilon \rightarrow 0$ proves the theorem. □

Remark 4.5. There is also a standard orthogonal-decomposition proof:

$$L^2 = \ker(U_T - I) \oplus \overline{\text{ran}(U_T - I)}.$$

On $\ker(U_T - I)$, the averages are unchanged. If $f = U_T g - g$, then

$$A_N f = \frac{U_T^N g - g}{N} \longrightarrow 0.$$

The result follows by density and the uniform bound $\|A_N\| \leq 1$.

4.4 Birkhoff's Pointwise Ergodic Theorem

Theorem 4.6 (Birkhoff). *Let (X, T) be an ergodic measure-preserving probability system and let $f \in L^1(X)$. Then*

$$\frac{1}{N} \sum_{n=0}^{N-1} f \circ T^n(x) \longrightarrow \mathbb{E}[f] \quad \text{for almost every } x.$$

The proof has two components. First, convergence is easy on a dense class. Functions of the form

$$f = h + (g - g \circ T), \quad h \circ T = h, \quad g \in L^\infty,$$

are dense in the relevant L^2 space, and

$$A_N f = h + \frac{g - g \circ T^N}{N} \longrightarrow h$$

pointwise. Second, the maximal inequality in the next subsection shows that the class of functions with almost-everywhere convergent averages is closed in L^1 .

To see the closure argument, suppose $f_k \rightarrow f$ in L^1 and each $A_N f_k$ converges almost everywhere. Define the oscillation

$$\Omega f(x) = \limsup_N A_N f(x) - \liminf_N A_N f(x) \geq 0.$$

The oscillation is subadditive, and $\Omega f_k = 0$, so

$$\Omega f \leq \Omega(f - f_k) \leq 2(f - f_k)^*.$$

The maximal inequality gives

$$\mu\{\Omega f > \alpha\} \leq \mu\{(f - f_k)^* > \alpha/2\} \leq \frac{2A}{\alpha} \|f - f_k\|_1 \longrightarrow 0.$$

Thus $\Omega f = 0$ almost everywhere. The value of the limit is identified as $\mathbb{E}[f]$ using the von Neumann theorem and an approximation argument.

4.5 The Maximal Ergodic Estimate

Define

$$M_T f(x) = \sup_{N \geq 1} \frac{1}{N} \sum_{n=0}^{N-1} |f \circ T^n(x)|.$$

Theorem 4.7 (Maximal ergodic weak-type estimate). *Let (X, μ) be a σ -finite measure space, let T be measure preserving, and let $f \in L^1(X)$. Then $M_T f(x) < \infty$ for almost every x , and*

there is a universal constant $A > 0$ such that

$$\mu\{x : M_T f(x) > \alpha\} \leq \frac{A}{\alpha} \|f\|_{L^1} \quad (\alpha > 0).$$

It is enough to consider $f \geq 0$.

Step 1: The Case $X = \mathbb{Z}$

Let $X = \mathbb{Z}$ with counting measure, $T(n) = n + 1$, and $f \geq 0$. Embed the discrete problem into $\mathbb{R}$ by defining

$$\tilde{f}(x) = f(n) \quad \text{when } n \leq x < n + 1.$$

If $E \subseteq \mathbb{Z}$, define

$$\tilde{E} = \bigcup_{n \in E} [n, n + 1).$$

Then

$$|\tilde{E}| = \#E, \quad \int_{\mathbb{R}} \tilde{f}(x) dx = \sum_{n \in \mathbb{Z}} f(n) = \|f\|_{\ell^1}.$$

Moreover,

$$\sum_{k=0}^{N-1} f(n+k) = \int_0^N \tilde{f}(n+t) dt.$$

If $x \in [n, n+1)$, then $[n, n+N] \subseteq [x-1, x+N]$, and therefore

$$\int_0^N \tilde{f}(n+t) dt \leq \int_{-1}^N \tilde{f}(x+t) dt.$$

Since $N \geq 1$, this yields

$$\frac{1}{N} \sum_{k=0}^{N-1} f(n+k) \leq \frac{N+1}{N} \frac{1}{N+1} \int_{-1}^N \tilde{f}(x+t) dt \leq 2 \tilde{f}^{*,\text{HL}}(x),$$

where $\tilde{f}^{*,\text{HL}}$ is the one-dimensional uncentered Hardy–Littlewood maximal function. Taking the supremum over N ,

$$\boxed{M_T f(n) \leq 2 \tilde{f}^{*,\text{HL}}(x) \quad \text{for every } x \in [n, n+1)}.$$

Consequently,

$$\begin{aligned} \#\{n : M_T f(n) > \alpha\} &\leq \left| \left\{ x : \tilde{f}^{*,\text{HL}}(x) > \frac{\alpha}{2} \right\} \right| \\ &\leq \frac{6}{\alpha} \|f\|_{\ell^1(\mathbb{Z})}. \end{aligned}$$

Thus the discrete estimate holds with $A = 6$.

Step 2: Transference from $\mathbb{Z}$ to X

For each N , define the truncated maximal function

$$M_{T,N} f(x) = \sup_{1 \leq m \leq N} \frac{1}{m} \sum_{k=0}^{m-1} f \circ T^k(x).$$

Since $M_{T,N}f \nearrow M_Tf$, it suffices to prove

$$\mu\{M_{T,N}f > \alpha\} \leq \frac{A}{\alpha} \|f\|_{L^1}$$

with A independent of N .

Fix N and write $M_Tf = M_{T,N}f$ temporarily. On $\mathbb{Z}$, set

$$M(F)(n) = \sup_{m \geq 1} \frac{1}{m} \sum_{k=0}^{m-1} F(n+k).$$

Encode each orbit in $X \times \mathbb{Z}$ by

$$F(x, n) = \begin{cases} f \circ T^n(x), & n \geq 0, \\ 0, & n < 0. \end{cases}$$

Then

$$A_m f(T^n x) = \frac{1}{m} \sum_{k=0}^{m-1} F(x, n+k).$$

Choose a positive integer a , let $B = a + N$, and truncate once more:

$$F_B(x, n) = \begin{cases} F(x, n), & n < B, \\ 0, & n \geq B. \end{cases}$$

If $m \leq N$ and $0 \leq n < a$, then $n+k < B$ for $0 \leq k < m$. Hence

$$A_m f(T^n x) \leq M(F_B(x, \cdot))(n),$$

and therefore

$$\boxed{M_T f(T^n x) \leq M(F_B(x, \cdot))(n) \quad \text{for } 0 \leq n < a.}$$

Let $E_\alpha = \{x : M_T f(x) > \alpha\}$. Measure preservation gives

$$\mu(E_\alpha) = \mu\{x : M_T f(T^n x) > \alpha\}$$

for every n . Summing over $0 \leq n < a$, applying Fubini–Tonelli, and then using the discrete estimate,

$$\begin{aligned} a\mu(E_\alpha) &= \int_X \#\{0 \leq n < a : M_T f(T^n x) > \alpha\} d\mu \\ &\leq \int_X \#\{n \in \mathbb{Z} : M(F_B(x, \cdot))(n) > \alpha\} d\mu \\ &\leq \int_X \frac{A}{\alpha} \|F_B(x, \cdot)\|_{\ell^1(\mathbb{Z})} d\mu \\ &= \frac{A}{\alpha} \int_X \sum_{k=0}^{B-1} f \circ T^k(x) d\mu \\ &= \frac{A}{\alpha} (a + N) \|f\|_{L^1}. \end{aligned}$$

Dividing by a and letting $a \rightarrow \infty$ gives

$$\mu(E_\alpha) \leq \frac{A}{\alpha} \|f\|_{L^1}.$$

The constant $A = 6$ is independent of N . Letting $N \rightarrow \infty$ proves the full maximal estimate.

4.6 Structural Summary

The main implications can be summarized as follows:

- The Vitali covering lemma implies the Hardy–Littlewood weak type $(1, 1)$ estimate, which implies Lebesgue differentiation.
- The Hardy–Littlewood weak estimate and Marcinkiewicz interpolation imply strong L^p bounds.
- The discrete maximal estimate and transference imply the maximal ergodic estimate.
- The maximal ergodic estimate and von Neumann’s L^2 convergence imply Birkhoff’s pointwise theorem.

The common principle is

maximal inequality + convergence on a dense subclass $\implies$ almost-everywhere convergence
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Part II

Weighted Estimates and Singular Integrals

5 Weighted Maximal Function Theory

5.1 Review and Motivation

Recall the uncentered Hardy–Littlewood maximal function

$$Mf(x) = \sup_{x \in B} \frac{1}{|B|} \int_B |f(y)| dy.$$

It satisfies

$$|\{Mf > \lambda\}| \leq \frac{C}{\lambda} \|f\|_{L^1}, \quad \|Mf\|_\infty \leq \|f\|_\infty,$$

and Marcinkiewicz interpolation consequently gives

$$\|Mf\|_p \leq C_p \|f\|_p, \quad 1 < p \leq \infty.$$

The argument extends in several directions: balls may be replaced by cubes, Lebesgue measure may be replaced by a doubling measure, and $\mathbb{R}^d$ may be replaced by a suitable metric space.

Definition 5.1. A measure μ on $\mathbb{R}^d$ is *doubling* if there is a constant C such that

$$\mu(2B) \leq C\mu(B)$$

for every ball B , where $2B$ is the concentric ball with twice the radius.

For a doubling measure, one may define

$$M_\mu f(x) = \sup_{x \in B} \frac{1}{\mu(B)} \int_B |f(y)| d\mu(y).$$

In this section, however, M always denotes the maximal operator whose averages are taken with respect to Lebesgue measure. We ask when this operator is bounded on a weighted space. Let

$$d\mu(x) = w(x) dx, \quad w \geq 0,$$

and use the notation

$$w(E) = \int_E w(x) dx, \quad f_B = \frac{1}{|B|} \int_B f(x) dx.$$

The problem is to characterize those w for which

$$M : L^p(w) \longrightarrow L^p(w).$$

5.2 Muckenhoupt's Theorem

Theorem 5.2 (Muckenhoupt). *Let $1 < p < \infty$ and $w \geq 0$. The following statements are equivalent:*

1. *There is $C > 0$ such that*

$$\|Mf\|_{L^p(w)} \leq C \|f\|_{L^p(w)}.$$

2. *There is $C > 0$ such that*

$$w(\{Mf > \lambda\}) \leq C \frac{\|f\|_{L^p(w)}^p}{\lambda^p}.$$

3. *There is $C > 0$ such that, for every ball B and every $f \geq 0$,*

$$\left(\frac{f(B)}{|B|} \right)^p \leq \frac{C}{w(B)} \int_B f^p w dx, \quad f(B) = \int_B f dx.$$

4. *The A_p characteristic*

$$[w]_{A_p} := \sup_B \left(\frac{1}{|B|} \int_B w dx \right) \left(\frac{1}{|B|} \int_B w^{-p'/p} dx \right)^{p/p'}$$

is finite, where $1/p + 1/p' = 1$.

Definition 5.3. We write $w \in A_p$ if $[w]_{A_p} < \infty$.

Applying the third condition to $f = \mathbf{1}_E$, where $E \subseteq B$, gives

$$\left(\frac{|E|}{|B|} \right)^p \leq C \frac{w(E)}{w(B)}.$$

In particular, every A_p weight is doubling. Indeed, take $E = \frac{1}{2}B$ to obtain

$$2^{-dp} \leq C \frac{w(\frac{1}{2}B)}{w(B)},$$

which is equivalent, after rescaling, to $w(2B) \leq C'w(B)$.

5.3 The Elementary Implications

Strong Type Implies Weak Type

Chebyshev's inequality gives

$$\lambda^p w(\{Mf > \lambda\}) \leq \int_{\{Mf > \lambda\}} (Mf)^p w \, dx \leq \|Mf\|_{L^p(w)}^p \leq C \|f\|_{L^p(w)}^p.$$

Weak Type Implies the Local Averaging Inequality

Fix a ball B , and replace f by $f\mathbf{1}_B$ if necessary. For every $x \in B$,

$$Mf(x) \geq \frac{1}{|B|} \int_B |f|.$$

If $\lambda < f(B)/|B|$, then $B \subseteq \{Mf > \lambda\}$, and hence

$$w(B) \leq w(\{Mf > \lambda\}) \leq C \frac{\|f\|_{L^p(w)}^p}{\lambda^p}.$$

Letting $\lambda \uparrow f(B)/|B|$ yields

$$\left(\frac{f(B)}{|B|} \right)^p \leq \frac{C}{w(B)} \int_B f^p w \, dx.$$

The Local Averaging Inequality Implies A_p

Take $f = w^{-p'/p}$. More precisely, first use

$$f = (w + \delta)^{-p'/p} \mathbf{1}_B$$

and then let $\delta \downarrow 0$. Since $p'/p = p' - 1$,

$$f^p w = w^{-p'} w = w^{1-p'} = w^{-p'/p} = f.$$

The local averaging inequality gives

$$\left(\frac{1}{|B|} \int_B w^{-p'/p} \right)^p \leq \frac{C}{w(B)} \int_B w^{-p'/p}.$$

Multiplying by $w(B)$ and dividing by the final integral,

$$|B|^{-p} \left(\int_B w^{-p'/p} \right)^{p-1} w(B) \leq C.$$

Since $p - 1 = p/p'$, this is precisely the A_p condition.

A_p Implies the Local Averaging Inequality

Hölder's inequality gives

$$f(B) = \int_B f w^{1/p} w^{-1/p} \leq \left(\int_B f^p w \right)^{1/p} \left(\int_B w^{-p'/p} \right)^{1/p'}.$$

Raising to the p -th power and dividing by $|B|^p$,

$$\left(\frac{f(B)}{|B|}\right)^p \leq \frac{1}{|B|^p} \left(\int_B f^p w\right) \left(\int_B w^{-p'/p}\right)^{p/p'}.$$

Because $1 + p/p' = p$, the A_p condition implies

$$\frac{w(B)}{|B|^p} \left(\int_B w^{-p'/p}\right)^{p/p'} \leq [w]_{A_p},$$

which proves the desired inequality.

The difficult implication is $w \in A_p \Rightarrow M : L^p(w) \rightarrow L^p(w)$. We prove it first for the dyadic maximal operator. We shall use the dual weight

$$\sigma = w^{-p'/p}.$$

Directly interchanging the two factors in the A_p condition shows that

$$w \in A_p \implies \sigma \in A_{p'}, \quad [\sigma]_{A_{p'}} = [w]_{A_p}^{p'/p}.$$

5.4 Dyadic Cubes and the Dyadic Maximal Operator

Let

$$\mathcal{D} = \left\{ 2^k \ell + [0, 2^k)^d : (\ell, k) \in \mathbb{Z}^d \times \mathbb{Z} \right\}$$

be the standard dyadic grid, and define

$$M_{\text{dyad}} f(x) = \sup_{\substack{Q \ni x \\ Q \in \mathcal{D}}} \frac{1}{|Q|} \int_Q |f(y)| dy.$$

If $Q_1, Q_2 \in \mathcal{D}$ intersect, then one contains the other. The dyadic parent $\pi_{\mathcal{D}}(Q)$ is the smallest dyadic cube strictly containing Q , and

$$|\pi_{\mathcal{D}}(Q)| = 2^d |Q|.$$

5.5 A Dyadic Calderón–Zygmund Decomposition

Lemma 5.4. *Let $f \in L^1(\mathbb{R}^d)$ and $\lambda > 0$. There is a pairwise disjoint family $\mathcal{B}$ of dyadic cubes such that*

(1)

$$\lambda < \frac{1}{|Q|} \int_Q |f| \leq 2^d \lambda \quad (Q \in \mathcal{B});$$

(2)

$$\sum_{Q \in \mathcal{B}} |Q| \leq \frac{1}{\lambda} \|f\|_{L^1};$$

(3)

$$\{M_{\text{dyad}} f > \lambda\} = \bigcup_{Q \in \mathcal{B}} Q.$$

Moreover, if

$$g = f \mathbf{1}_{\left(\bigcup_{Q \in \mathcal{B}} Q\right)^c}, \quad b = f - g = \sum_{Q \in \mathcal{B}} f \mathbf{1}_Q,$$

then $|g| \leq \lambda$ almost everywhere.

Proof. Let $\mathcal{B}$ be the family of maximal dyadic cubes satisfying

$$\frac{1}{|Q|} \int_Q |f| > \lambda.$$

Such maximal cubes exist because

$$\frac{1}{|Q|} \int_Q |f| \leq \frac{\|f\|_1}{|Q|} \rightarrow 0$$

as the side length tends to infinity. Maximality and the nesting property imply that the cubes in $\mathcal{B}$ are pairwise disjoint.

For $Q \in \mathcal{B}$, its parent does not satisfy the defining inequality. Therefore

$$\lambda \geq \frac{1}{|\pi_{\mathcal{D}}(Q)|} \int_{\pi_{\mathcal{D}}(Q)} |f| \geq \frac{1}{2^d |Q|} \int_Q |f|,$$

which proves the upper bound in (1).

Every bad cube lies in the superlevel set in (3). Conversely, if $M_{\text{dyad}} f(x) > \lambda$, then a dyadic cube containing x satisfies the bad-average condition and is contained in a maximal such cube. This proves (3). Finally, disjointness and the lower bound in (1) give

$$\sum_{Q \in \mathcal{B}} |Q| \leq \frac{1}{\lambda} \sum_{Q \in \mathcal{B}} \int_Q |f| \leq \frac{1}{\lambda} \|f\|_1.$$

Outside the union of the bad cubes, every dyadic average containing x is at most λ . The Lebesgue differentiation theorem then gives $|f(x)| \leq \lambda$ almost everywhere there. $\square$

5.6 Weighted Boundedness of the Dyadic Maximal Operator

Theorem 5.5. *If $1 < p < \infty$ and $w \in A_p$, then*

$$M_{\text{dyad}} : L^p(w) \longrightarrow L^p(w)$$

is bounded.

Proof. Choose $c_0 > 2^{d+1} + 1$. For each $k \in \mathbb{Z}$, let $\mathcal{B}_k$ be the family of bad cubes from the preceding decomposition at height c_0^k , and set

$$\mathcal{B} = \bigcup_{k \in \mathbb{Z}} \mathcal{B}_k.$$

For $Q \in \mathcal{B}$, define

$$E(Q) = Q \setminus \bigcup_{\substack{Q' \in \mathcal{B} \\ Q' \subsetneq Q}} Q'.$$

We first claim that

$$|E(Q)| \geq \frac{1}{2} |Q|.$$

Suppose $Q \in \mathcal{B}_k$. For $j \leq k$, the cube Q lies in a maximal cube of $\mathcal{B}_j$, so no member of $\mathcal{B}_j$ is strictly contained in Q . For $j \geq k+1$, disjointness within $\mathcal{B}_j$ and the Calderón–Zygmund bounds give

$$\sum_{\substack{Q' \in \mathcal{B}_j \\ Q' \subseteq Q}} |Q'| \leq \frac{1}{c_0^j} \int_Q |f| \leq \frac{2^d c_0^k}{c_0^j} |Q|.$$

Summing over $j \geq k+1$,

$$\left| \bigcup_{\substack{Q' \in \mathcal{B} \\ Q' \subsetneq Q}} Q' \right| \leq 2^d |Q| \sum_{j \geq k+1} c_0^{k-j} = 2^d |Q| \frac{c_0^{-1}}{1 - c_0^{-1}} \leq \frac{1}{2} |Q|.$$

This proves the claim. The sets $E(Q)$ are pairwise disjoint.

If $x \in E(Q)$ with $Q \in \mathcal{B}_k$, then x belongs to no cube in $\mathcal{B}_{k+1}$, and therefore

$$M_{\text{dyad}} f(x) \leq c_0^{k+1}.$$

Consequently,

$$\begin{aligned} \int |M_{\text{dyad}} f|^p w \, dx &\leq \sum_k \sum_{Q \in \mathcal{B}_k} c_0^{(k+1)p} w(E(Q)) \\ &\leq C_p \sum_k \sum_{Q \in \mathcal{B}_k} \left(\frac{1}{|Q|} \int_Q |f| \right)^p w(E(Q)). \end{aligned}$$

Introduce $\sigma = w^{-p'/p}$. We rewrite

$$\frac{1}{|Q|} \int_Q |f| = \left(\frac{1}{\sigma(Q)} \int_Q |f| \sigma^{-1} \sigma \, dy \right) \frac{\sigma(Q)}{|Q|}.$$

Thus the preceding sum equals

$$\begin{aligned} \sum_{Q \in \mathcal{B}} \left(\frac{1}{\sigma(Q)} \int_Q |f| \sigma^{-1} \sigma \right)^p \sigma(E(Q)) \\ \times \underbrace{\frac{w(E(Q))}{\sigma(E(Q))} \left(\frac{\sigma(Q)}{|Q|} \right)^p}_I. \end{aligned}$$

Since $\sigma \in A_{p'}$, the set estimate for $A_{p'}$ weights and $|E(Q)| \geq |Q|/2$ imply

$$\sigma(E(Q)) \geq c \sigma(Q).$$

Also $w(E(Q)) \leq w(Q)$, so

$$\begin{aligned} I &\leq C \frac{w(Q)}{\sigma(Q)} \frac{\sigma(Q)^p}{|Q|^p} \\ &= C \left(\frac{1}{|Q|} \int_Q w \right) \left(\frac{1}{|Q|} \int_Q \sigma \right)^{p-1} \\ &\leq C[w]_{A_p}. \end{aligned}$$

For $y \in E(Q) \subseteq Q$,

$$\frac{1}{\sigma(Q)} \int_Q |f\sigma^{-1}| \sigma \leq M_\sigma(f\sigma^{-1})(y).$$

Using the disjointness of the $E(Q)$ and the $L^p(\sigma)$ boundedness of M_σ , which holds because σ is doubling, we obtain

$$\begin{aligned} \int |M_{\text{dyad}} f|^p w &\leq C_{p,w} \sum_{Q \in \mathcal{B}} \int_{E(Q)} (M_\sigma(f\sigma^{-1}))^p \sigma \, dy \\ &\leq C_{p,w} \int_{\mathbb{R}^d} (M_\sigma(f\sigma^{-1}))^p \sigma \, dy \\ &\leq C_{p,w} \int_{\mathbb{R}^d} |f\sigma^{-1}|^p \sigma \, dy. \end{aligned}$$

Finally,

$$\sigma^{1-p} = w^{-\frac{p'}{p}(1-p)} = w,$$

and hence

$$\int |M_{\text{dyad}} f|^p w \leq C_{p,w} \int |f|^p w.$$

□

6 The Hilbert Transform

6.1 Singular Kernels and Singular Integrals

Definition 6.1. A kernel

$$K : \mathbb{R}^d \times \mathbb{R}^d \setminus \{(x, x) : x \in \mathbb{R}^d\} \longrightarrow \mathbb{C}$$

is singular if

$$|K(x, y)| \sim |x - y|^{-d} \quad \text{as } |x - y| \rightarrow 0.$$

A singular integral is formally an operator of the form

$$Tf(x) = \int K(x, y)f(y) \, dy.$$

Such operators occur throughout partial differential equations, ergodic theory, and signal processing.

6.2 Definition of the Hilbert Transform

For the Hilbert transform, the convolution kernel is

$$h(x) = \frac{1}{\pi x}, \quad K(x, y) = h(x - y).$$

Since $1/x \notin L^1_{\text{loc}}(\mathbb{R})$, the integral must be interpreted as a principal value.

Recall that the Schwartz space $\mathcal{S}(\mathbb{R})$ is the Fréchet space defined by the seminorms

$$\|f\|_{n,m} = \sup_{x \in \mathbb{R}} |x^m f^{(n)}(x)|.$$

Define the principal-value distribution w_0 by

$$\langle w_0, f \rangle = \frac{1}{\pi} \lim_{\varepsilon \rightarrow 0} \int_{\varepsilon \leq |x| \leq 1} \frac{f(x)}{x} dx + \frac{1}{\pi} \int_{|x| \geq 1} \frac{f(x)}{x} dx.$$

This is well defined because

$$\int_{\varepsilon \leq |x| \leq 1} \frac{f(x)}{x} dx = \int_{\varepsilon \leq |x| \leq 1} \frac{f(x) - f(0)}{x} dx,$$

and

$$\left| \frac{f(x) - f(0)}{x} \right| \leq \sup_x |f'(x)|.$$

The integral at infinity converges because Schwartz functions decay rapidly.

Definition 6.2. For $f \in L^p(\mathbb{R})$, $1 \leq p < \infty$, define the truncated Hilbert transform by

$$H^{(\varepsilon)} f(x) = \frac{1}{\pi} \int_{|y| \geq \varepsilon} \frac{f(x-y)}{y} dy = \frac{1}{\pi} \int_{|x-y| \geq \varepsilon} \frac{f(y)}{x-y} dy.$$

The truncated integral is well defined by Hölder's inequality. For $f \in \mathcal{S}(\mathbb{R})$,

$$Hf(x) = (w_0 * f)(x) = \lim_{\varepsilon \rightarrow 0} H^{(\varepsilon)} f(x).$$

6.3 The Fourier Multiplier

Proposition 6.3. *With the convention*

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx,$$

one has

$$\widehat{w_0}(\xi) = -i \operatorname{sgn}(\xi).$$

Proof. For $\varphi \in \mathcal{S}(\mathbb{R})$,

$$\begin{aligned} \langle \widehat{w_0}, \varphi \rangle &= \langle w_0, \widehat{\varphi} \rangle \\ &= \frac{1}{\pi} \lim_{\varepsilon \rightarrow 0} \int_{\varepsilon \leq |\xi| \leq 1/\varepsilon} \widehat{\varphi}(\xi) \frac{d\xi}{\xi} \\ &= \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}} \varphi(x) \left[\frac{1}{\pi} \int_{\varepsilon \leq |\xi| \leq 1/\varepsilon} e^{-2\pi i x \xi} \frac{d\xi}{\xi} \right] dx. \end{aligned}$$

The cosine term vanishes by oddness. Therefore the expression in brackets is

$$-\frac{i}{\pi} \int_{\varepsilon \leq |\xi| \leq 1/\varepsilon} \frac{\sin(2\pi x \xi)}{\xi} d\xi = -\frac{2i}{\pi} \operatorname{sgn}(x) \int_{2\pi \varepsilon |x|}^{2\pi |x|/\varepsilon} \frac{\sin u}{u} du.$$

The Dirichlet integral

$$\int_0^\infty \frac{\sin u}{u} du = \frac{\pi}{2}$$

shows that this converges pointwise for $x \neq 0$ to $-i \operatorname{sgn}(x)$. The truncated integrals are uniformly

bounded, so dominated convergence yields

$$\langle \widehat{w_0}, \varphi \rangle = \int_{\mathbb{R}} \varphi(x) (-i \operatorname{sgn}(x)) dx.$$

□

Thus

$$Hf = \left[-i \operatorname{sgn}(\xi) \widehat{f}(\xi) \right]^\vee.$$

Plancherel's theorem immediately gives

$$\|Hf\|_2 = \|f\|_2, \quad H^2 = -I, \quad H^* = -H.$$

6.4 L^p Boundedness

Theorem 6.4. *For every $1 < p < \infty$, the Hilbert transform is bounded on $L^p(\mathbb{R})$.*

Proof. By duality and $H^* = -H$, it suffices to treat $1 < p \leq 2$. Marcinkiewicz interpolation and the L^2 isometry reduce the problem to the weak-type estimate

$$|\{x : |Hf(x)| > \lambda\}| \leq \frac{C}{\lambda} \|f\|_1.$$

Perform a one-dimensional Calderón–Zygmund decomposition of f at height λ . There are pairwise disjoint dyadic intervals $\{I_j\}$ such that

$$\lambda < \frac{1}{|I_j|} \int_{I_j} |f| \leq 2\lambda.$$

Let $\Omega = \bigcup_j I_j$, and write $f = g + b$, where

$$g(x) = \begin{cases} f(x), & x \notin \Omega, \\ \frac{1}{|I_j|} \int_{I_j} f, & x \in I_j, \end{cases}$$

and

$$b = \sum_j b_j, \quad b_j = \left(f - \frac{1}{|I_j|} \int_{I_j} f \right) \mathbf{1}_{I_j}.$$

Then

$$\int_{I_j} b_j = 0, \quad |\Omega| \leq \frac{1}{\lambda} \|f\|_1.$$

For the good part, Chebyshev's inequality and the L^2 isometry give

$$|\{|Hg| > \lambda/2\}| \leq \frac{4}{\lambda^2} \|Hg\|_2^2 = \frac{4}{\lambda^2} \|g\|_2^2.$$

Outside Ω , one has $g = f$ and $|f| \leq \lambda$ almost everywhere; on each I_j , one has $|g| \leq 2\lambda$. Hence

$$\begin{aligned} \int |g|^2 &\leq \lambda \int_{\mathbb{R} \setminus \Omega} |f| + 4\lambda^2 |\Omega| \\ &\leq C\lambda \|f\|_1. \end{aligned}$$

It follows that

$$|\{|Hg| > \lambda/2\}| \leq \frac{C}{\lambda} \|f\|_1.$$

For the bad part, let $I_j^* = 2I_j$ and

$$E = \bigcup_j I_j^*.$$

Then

$$|E| \leq 2|\Omega| \leq \frac{2}{\lambda} \|f\|_1.$$

Let c_j be the center of I_j . If $x \notin I_j^*$, the cancellation $\int b_j = 0$ gives

$$\begin{aligned} |Hb_j(x)| &= \left| \frac{1}{\pi} \int_{I_j} \left(\frac{1}{x-y} - \frac{1}{x-c_j} \right) b_j(y) dy \right| \\ &\leq C \int_{I_j} \frac{|y-c_j|}{|x-c_j|^2} |b_j(y)| dy \\ &\leq C \frac{|I_j|}{|x-c_j|^2} \|b_j\|_1. \end{aligned}$$

Therefore, for $x \in E^c$,

$$|Hb(x)| \leq C \sum_j \frac{|I_j|}{|x-c_j|^2} \|b_j\|_1.$$

Chebyshev's inequality now gives

$$\begin{aligned} |\{x \notin E : |Hb(x)| > \lambda/2\}| &\leq \frac{2}{\lambda} \int_{\mathbb{R} \setminus E} |Hb(x)| dx \\ &\leq \frac{C}{\lambda} \sum_j \|b_j\|_1 \int_{|x-c_j| \geq |I_j|} \frac{|I_j|}{|x-c_j|^2} dx. \end{aligned}$$

Since

$$\int_{|x-c_j| \geq |I_j|} \frac{|I_j|}{|x-c_j|^2} dx = 2$$

and

$$\sum_j \|b_j\|_1 \leq 2 \|f\|_1,$$

we obtain

$$|\{x \notin E : |Hb(x)| > \lambda/2\}| \leq \frac{C}{\lambda} \|f\|_1.$$

Combining the good and bad estimates proves weak type $(1, 1)$, and interpolation and duality complete the proof. $\square$

7 The Fourier Transform on Euclidean Space

Throughout this section,

$$\widehat{f}(\xi) = \int_{\mathbb{R}^d} e^{-2\pi i x \cdot \xi} f(x) dx.$$

7.1 Convolution and the Multiplication Formula

Proposition 7.1. (1) If $f, g \in L^1(\mathbb{R}^d)$, then

$$\widehat{f * g}(\xi) = \widehat{f}(\xi)\widehat{g}(\xi).$$

(2) If $f, g \in \mathcal{S}(\mathbb{R}^d)$, then

$$\int_{\mathbb{R}^d} f(x)\widehat{g}(x) dx = \int_{\mathbb{R}^d} \widehat{f}(y)g(y) dy.$$

Proof. Absolute integrability permits Fubini's theorem:

$$\begin{aligned} \widehat{f * g}(\xi) &= \int e^{-2\pi i x \cdot \xi} \int f(x - y)g(y) dy dx \\ &= \int e^{-2\pi i y \cdot \xi} g(y) \int f(x - y)e^{-2\pi i (x - y) \cdot \xi} dx dy \\ &= \widehat{f}(\xi)\widehat{g}(\xi). \end{aligned}$$

Similarly,

$$\begin{aligned} \int f(x)\widehat{g}(x) dx &= \iint f(x)g(y)e^{-2\pi i y \cdot x} dy dx \\ &= \int \widehat{f}(y)g(y) dy. \end{aligned}$$

□

7.2 Basic Examples and Decay

In the sense of tempered distributions,

$$\widehat{\delta}_0 = 1.$$

If $f \in C_c^\infty(\mathbb{R}^d)$, then $\widehat{f} \in \mathcal{S}(\mathbb{R}^d)$, but $\widehat{f}$ is not compactly supported unless $f = 0$. Indeed, $\widehat{f}$ extends to an entire function. This is an elementary manifestation of the uncertainty principle: localization in one domain produces spreading in the other.

Proposition 7.2. If $f \in L^1(\mathbb{R}^d) \cap C_c^k(\mathbb{R}^d)$, then

$$|\widehat{f}(\xi)| \leq \frac{C_f}{\langle \xi \rangle^k}, \quad \langle \xi \rangle = (1 + |\xi|^2)^{1/2}.$$

Proof. Integration by parts in the first coordinate gives

$$\begin{aligned} \widehat{f}(\xi) &= \int f(x) \frac{\partial}{\partial x_1} \left(e^{-2\pi i x \cdot \xi} \right) \frac{1}{-2\pi i \xi_1} dx \\ &= \frac{1}{2\pi i \xi_1} \int \frac{\partial f}{\partial x_1}(x) e^{-2\pi i x \cdot \xi} dx. \end{aligned}$$

The boundary term vanishes because f is compactly supported. Iterate k times in a coordinate direction for which $|\xi_j|$ is maximal. Together with the trivial L^1 estimate for small ξ , this proves the result. □

7.3 The Gaussian

Proposition 7.3. *The Gaussian is fixed by the Fourier transform:*

$$\widehat{e^{-\pi|x|^2}}(\xi) = e^{-\pi|\xi|^2}.$$

Proof. The product structure and Fubini's theorem reduce the proof to one dimension. Completing the square,

$$-2\pi ix\xi - \pi x^2 = -\pi(x + i\xi)^2 - \pi\xi^2.$$

Thus

$$\widehat{e^{-\pi x^2}}(\xi) = e^{-\pi\xi^2} \int_{\mathbb{R}} e^{-\pi(x+i\xi)^2} dx.$$

The function $e^{-\pi z^2}$ is entire. Integrate it around the rectangle with vertices

$$-R, \quad R, \quad R + i\xi, \quad -R + i\xi.$$

The two vertical integrals have modulus at most

$$|\xi|e^{-\pi R^2 + \pi\xi^2},$$

which tends to zero as $R \rightarrow \infty$. Hence

$$\int_{\mathbb{R}} e^{-\pi(x+i\xi)^2} dx = \int_{\mathbb{R}} e^{-\pi x^2} dx = 1,$$

and the claim follows. □

7.4 Fourier Inversion on the Schwartz Space

Theorem 7.4. *The Fourier transform*

$$\mathcal{F} : \mathcal{S}(\mathbb{R}^d) \longrightarrow \mathcal{S}(\mathbb{R}^d)$$

is a topological isomorphism. Its inverse is

$$\mathcal{F}^{-1}f(x) = \int_{\mathbb{R}^d} e^{2\pi i x \cdot \xi} f(\xi) d\xi.$$

Proof. The formal computation

$$\mathcal{F}^{-1}\mathcal{F}f(x) = \int f(y) \left(\int e^{2\pi i(x-y) \cdot \xi} d\xi \right) dy$$

is invalid because the inner integral is not absolutely convergent. We regularize it with a Gaussian:

$$\mathcal{F}^{-1}\mathcal{F}f(x) = \lim_{\varepsilon \rightarrow 0} \int e^{2\pi i x \cdot \xi} \widehat{f}(\xi) e^{-\pi\varepsilon^2|\xi|^2} d\xi.$$

Since $\widehat{f} \in L^1$, dominated convergence justifies this limit. Fubini's theorem is now applicable:

$$\mathcal{F}^{-1}\mathcal{F}f(x) = \lim_{\varepsilon \rightarrow 0} \int f(y) \int e^{2\pi i(x-y) \cdot \xi} e^{-\pi\varepsilon^2|\xi|^2} d\xi dy.$$

The scaling law for the Gaussian gives

$$\int e^{2\pi i(x-y)\cdot\xi} e^{-\pi\varepsilon^2|\xi|^2} d\xi = \frac{1}{\varepsilon^d} e^{-\pi|(y-x)/\varepsilon|^2} =: G_\varepsilon(x-y).$$

The family $\{G_\varepsilon\}$ is an approximate identity:

$$G_\varepsilon \geq 0, \quad \int G_\varepsilon = 1,$$

and its mass concentrates at the origin. Therefore

$$\mathcal{F}^{-1}\mathcal{F}f(x) = \lim_{\varepsilon \rightarrow 0} \int f(y) G_\varepsilon(x-y) dy = f(x).$$

The standard seminorm estimates show continuity of $\mathcal{F}$ and its inverse on the Schwartz space. $\square$

7.5 Plancherel's Theorem

Theorem 7.5 (Plancherel). *The Fourier transform extends uniquely to a unitary operator on $L^2(\mathbb{R}^d)$. In particular,*

$$\|\widehat{f}\|_2 = \|f\|_2, \quad \langle \widehat{f}, \widehat{g} \rangle_{L^2} = \langle f, g \rangle_{L^2}.$$

Proof. First let $f \in \mathcal{S}$, and set

$$g = \mathcal{F}^{-1}(\overline{f}).$$

Then

$$g(x) = \int e^{2\pi i x \cdot \xi} \overline{f(\xi)} d\xi = \overline{\widehat{f}(x)},$$

and $\widehat{g} = \overline{f}$. Applying the multiplication formula,

$$\|f\|_2^2 = \int f \overline{f} = \int f \widehat{g} = \int \widehat{f} g = \int |\widehat{f}|^2 = \|\widehat{f}\|_2^2.$$

The polarization identity gives preservation of inner products. Since $\mathcal{S}$ is dense in L^2 , $\mathcal{F}$ extends uniquely to an isometry on L^2 . Fourier inversion on $\mathcal{S}$ implies that its range is dense and closed, hence all of L^2 . $\square$

7.6 The Hausdorff–Young Theorem

Theorem 7.6 (Hausdorff–Young). *If $1 \leq p \leq 2$, then the Fourier transform extends to a bounded operator*

$$\mathcal{F} : L^p(\mathbb{R}^d) \longrightarrow L^{p'}(\mathbb{R}^d), \quad \frac{1}{p} + \frac{1}{p'} = 1.$$

Proof. At the endpoints,

$$\mathcal{F} : L^1 \rightarrow L^\infty, \quad \|\widehat{f}\|_\infty \leq \|f\|_1,$$

and Plancherel gives

$$\mathcal{F} : L^2 \rightarrow L^2$$

isometrically. Riesz–Thorin interpolation gives

$$\frac{1}{p} = \frac{1-\theta}{1} + \frac{\theta}{2}, \quad \frac{1}{q} = \frac{1-\theta}{\infty} + \frac{\theta}{2} = \frac{\theta}{2}.$$

Thus $1/p + 1/q = 1$, so $q = p'$. $\square$

Finally, the Fourier transform extends to the space of tempered distributions $\mathcal{S}'$ by duality:

$$\langle \widehat{T}, \varphi \rangle = \langle T, \widehat{\varphi} \rangle.$$

Part III

Randomization and Pointwise Fourier Convergence

8 Probability and Randomization

8.1 Basic probability theory

Probability space $(\Omega, \Sigma, \mathbb{P})$, with $A \in \Sigma$ the *events*.

INDEPENDENCE.

- A, B are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$.
- Let $\Sigma_1, \Sigma_2 \subseteq \Sigma$ be two σ -subalgebras. Σ_1, Σ_2 are independent if for all $A \in \Sigma_1, B \in \Sigma_2$, A and B are independent.

RANDOM VARIABLES.

- A random variable is a measurable map $X : \Omega \rightarrow \mathbb{R}$.
- $\sigma(X) \subseteq \Sigma$ is the smallest σ -algebra making X measurable.
- X_1, X_2 are independent random variables if $\sigma(X_1), \sigma(X_2)$ are independent.

EXAMPLE (RADEMACHER FUNCTIONS). $\Omega = [0, 1]$, Σ = Borel algebra, $\mathbb{P}$ = Lebesgue measure. For $j \geq 0$, define $r_j : \Omega \rightarrow \mathbb{R}$ by

$$r_j(t) = \operatorname{sgn}(\sin(2\pi 2^j t)).$$

These are independent, and

$$\mathbb{P}((r_{j_i} = \varepsilon_i)_{i=1}^n) = 2^{-n}, \quad \varepsilon_i \in \{1, -1\}.$$

8.2 The Borel–Cantelli lemmas

Let $(A_j)_{j=1}^\infty$ be an infinite sequence in a probability space $(\Omega, \Sigma, \mathbb{P})$, and consider $\sum_j \mathbf{1}_{A_j}$.

If

$$\mathbb{E}\left(\sum_j \mathbf{1}_{A_j}\right) = \sum_j \mathbb{E}(\mathbf{1}_{A_j}) = \sum_j \mathbb{P}(A_j) < +\infty,$$

then $\sum_j \mathbf{1}_{A_j} < +\infty$ a.e. Note that

$$\{x \mid x \in A_j \text{ for infinitely many } j\} = \bigcap_k \bigcup_{j>k} A_j.$$

LEMMA.

1. $\sum_j \mathbb{P}(A_j) < +\infty \implies \mathbb{P}(A_j \text{ occurs infinitely often}) = 0.$
2. If $\{A_j\}$ are independent and $\sum_j \mathbb{P}(A_j) = +\infty$, then $\mathbb{P}(\bigcap_k \bigcup_{j>k} A_j) = 1.$

PROOF. Use $1 - x \leq e^{-x}$ for all $x \geq 0$. Then

$$\mathbb{P}\left(\bigcap_{j>k}^N A_j^c\right) = \prod_{j>k}^N (1 - \mathbb{P}(A_j)) \leq e^{-\sum \mathbb{P}(A_j)} \xrightarrow{N \rightarrow +\infty} 0.$$

Hence $\mathbb{P}(\bigcup_{j>k}^{+\infty} A_j) = 1$, which gives the claim. $\square$

8.3 Kolmogorov's zero-one law

THEOREM. Let X_n be independent. Then for all $A \in \mathcal{F}_\infty$ we have $\mathbb{P}(A) \in \{0, 1\}$, where

$$\mathcal{F}_\infty = \bigcap_{n=1}^{\infty} \sigma(X_n, X_{n+1}, \dots) \quad (\text{the tail } \sigma\text{-algebra / tail events}).$$

PROOF. Put $G_n = \sigma(X_1, \dots, X_n)$, which is independent of $\sigma(X_{n+1}, \dots)$. Hence any $A \in \mathcal{F}_\infty$ is independent of $\bigcup_n G_n$, i.e. $\mathcal{F}_\infty$ is independent of every G_n . Since $\mathcal{F}_\infty \subseteq \sigma(\bigcup_n G_n)$, we get

$$\mathbb{P}(A) = \mathbb{P}(A \cap A) = \mathbb{P}(A)^2. \quad \square$$

8.4 Sub-Gaussian bounds

Setting: X_i i.i.d. with $\mathbb{P}\{X_i = 1\} = \frac{1}{2}$.

LEMMA. Let $(r_j)_{j=1}^{+\infty}$ be a sequence of Rademacher functions. For all $N \geq 1$ and $(a_j)_{j=1}^N \subseteq \mathbb{C}^N$, set

$$S_N = \sum_{j=1}^N a_j r_j.$$

For all $\lambda > 0$: since $\mathbb{E}[r_j] = 0$, we have

$$\text{Var}(S_N) = \mathbb{E}[|S_N|^2] = \|a\|_{\ell^2(\mathbb{R}^N)}^2 = \sigma^2,$$

and

$$\mathbb{P}(|S_N| > \lambda \sigma) \leq 4e^{-\lambda^2/4}.$$

This measures how far S_N can be from its expectation.

PROOF. Suppose first $a_j \in \mathbb{R}$, $j = 1, \dots, N$. Let $t > 0$. By independence,

$$\mathbb{E} e^{tS_N} = \prod_{j=1}^N \mathbb{E}(e^{tr_j a_j}) = \prod_j \frac{e^{ta_j} + e^{-ta_j}}{2} = \prod_j \cosh(ta_j).$$

Using the elementary inequality

$\cosh(x) \leq e^{x^2/2}$

we obtain

$$\mathbb{E} e^{tS_N} \leq \prod_j e^{\frac{1}{2}(ta_j)^2} = e^{t^2(\sum_j a_j^2)/2} = e^{\frac{t^2}{2}\sigma^2}.$$

Therefore, by Chebyshev's inequality applied to the exponential,

$$\mathbb{P}(S_N > \lambda\sigma) = \mathbb{P}(e^{tS_N} > e^{t\lambda\sigma}) \leq e^{-t\lambda\sigma} \mathbb{E} e^{tS_N} = e^{-t\lambda\sigma + \frac{1}{2}t^2\sigma^2}.$$

Choosing $t = \frac{\lambda}{\sigma}$ gives $e^{-\lambda^2/2}$. The same argument applies to $S_N < -\lambda\sigma$, hence

$$\mathbb{P}(|S_N| > \lambda\sigma) \leq 2e^{-\lambda^2/2}.$$

COMPLEX CASE. $|S_N|^2 = |\operatorname{Re}(S_N)|^2 + |\operatorname{Im}(S_N)|^2 \geq \lambda^2\sigma^2$ forces

$$|\operatorname{Re}(S_N)|^2 \geq \frac{\lambda^2\sigma^2}{2} \quad \text{or} \quad |\operatorname{Im}(S_N)|^2 \geq \frac{\lambda^2\sigma^2}{2},$$

so the bound becomes $\leq 4e^{-\lambda^2/4}$. $\square$

8.5 Khinchine's inequality

THEOREM (KHINCHINE). For $1 \leq p < +\infty$ there exists $c = c(p) > 0$ such that

$$\frac{1}{c} \|a\|_2 \leq \|S_N\|_p \leq c \|a\|_2.$$

PROOF.

Case $p = 2$: $\|S_N\|_{L^2} = \|a\|_{\ell^2}$. WLOG $\|a\|_{\ell^2} = 1 = \sigma$.

Upper bound:

$$\mathbb{E}(|S_N|^p) = \int_0^\infty \mathbb{P}(|S_N| > \lambda) p \lambda^{p-1} d\lambda \leq \int_0^{+\infty} 4e^{-\lambda^2/4} p \lambda^{p-1} d\lambda \leq c(p) < +\infty.$$

Lower bound: Since $p \mapsto \|f\|_p$ is increasing for $p \geq 1$, it suffices to prove the case $p = 1$. Write

$$\begin{aligned} \int |S_N|^2 &= \int |S_N|^{2/3} |S_N|^{4/3} \\ &\leq \left(\int |S_N| \right)^{2/3} \left(\int |S_N|^4 \right)^{1/3} \\ &\leq C \left(\int |S_N| \right)^{2/3} \left(\int |S_N|^2 \right)^{2/3}, \end{aligned}$$

where the last step used the upper bound for $p = 4$. Hence

$$\frac{1}{c} \int |S_N|^2 \leq \left(\int |S_N| \right)^2. \quad \square$$

9 Fourier Partial Sums and Divergence

9.1 Partial sums

For $f \in L^1(\mathbb{T})$ and $N \geq 1$, let $D_N = \sum_{|n| \leq N} e(nx)$ be the Dirichlet kernel. Then

$$S_N f(x) = \sum_{n=-N}^N \widehat{f}(n) e^{-2\pi i n x} = D_N * f(x).$$

The central question is whether $S_N f$ converges to f almost everywhere.

- **1968: Carleson** — $f \in L^2$, via time-frequency analysis.
- **Hunt** — $f \in L^p$.
- Open-ended problem: $\lim_{r \rightarrow +\infty} \int_{B_r} \widehat{f}(\xi) e^{2\pi i x \xi} d\xi \xrightarrow{\text{a.e.}} f(x)$ — higher-order Fourier analysis.

9.2 Norm convergence and uniform boundedness

PROPOSITION. Let $1 \leq p \leq \infty$. The following statements are equivalent:

1. For all $f \in L^p(\mathbb{T})$, $\|S_N f - f\|_p \rightarrow 0$;
2. $\sup_N \|S_N\|_{p \rightarrow p} < +\infty$.

PROOF.

(ii) $\Rightarrow$ (i): Take $g \in C_c^\infty$ and split

$$\|S_N f - f\|_p \leq \underbrace{\|S_N f - S_N g\|_p}_{\text{bounded operator}} + \underbrace{\|S_N g - g\|_p}_{\rightarrow 0} + \underbrace{\|g - f\|_p}_{< \varepsilon/3}.$$

(i) $\Rightarrow$ (ii): Uniform boundedness principle. $\square$

THEOREM. For $1 < p < +\infty$, $\sup_N \|S_N\|_{p \rightarrow p} < +\infty$.

PROOF. The Hilbert transform on $\mathbb{T}$ is the multiplier

$$\widehat{Hf}(n) = -i \operatorname{sgn}(n) \widehat{f}(n),$$

so that

$$Hf = \sum_{n>0} -i \widehat{f}(n) + \sum_{n<0} i \widehat{f}(n) = \sum_{n \geq 0} -i \widehat{f}(n) + \sum_{n \leq 0} i \widehat{f}(n)$$

(the added terms $-i \widehat{f}(0) + i \widehat{f}(0)$ cancel). Write $e_n(x) = e(nx)$. Then

$$Tf := \frac{1}{2}(1 + iH)(f) = \frac{1}{2}f + \frac{i}{2}Hf = \sum_{n \geq 0} \widehat{f}(n) e_n - \frac{1}{2}\widehat{f}(0),$$

i.e. T is the Riesz projection onto nonnegative frequencies. Using

$$\mathbf{1}_{[-N, N]} = \mathbf{1}_{[-N, \infty)} - \mathbf{1}_{(N, \infty)},$$

one gets

$$S_N f = \bar{e}_N T(e_N f) - e_N T(\bar{e}_N f) - \frac{1}{2}(\widehat{f}(-N)e_{-N} + \widehat{f}(N)e_N),$$

where the last (rank-two) term has L^p norm $\leq \|f\|_p$. Hence

$$\|S_N\|_{p \rightarrow p} \leq 2\|T\|_{p \rightarrow p} \leq 1 + \|H\|_{p \rightarrow p}. \quad \square$$

MAXIMAL OPERATOR. Define

$$Cf(x) = \sup_N |S_N f(x)|.$$

9.3 Kolmogorov's divergence theorem

THEOREM (KOLMOGOROV). There exists $f \in L^1(\mathbb{T})$ with $S_N f \not\rightarrow f$ a.e.; in fact there exists $f \in L^1(\mathbb{T})$ such that

$$|\{x \in \mathbb{T} \mid Cf(x) < \infty\}| = 0.$$

COROLLARY. If $S_N f \rightarrow f$ a.e. for all $f \in L^1$, then there exists $A > 0$ with

$$|\{x \in \mathbb{T} \mid C\mu(x) > \lambda\}| \leq \frac{A}{\lambda} \|\mu\| = \frac{A}{\lambda} |\mu|(\mathbb{T}),$$

for μ a complex finite Borel measure.

LEMMA. There exist probability measures μ_N on $\mathbb{T}$ with

$$\limsup_n |S_n \mu_N(x)| \gtrsim \log(N) \quad \text{a.e.}$$

Combining: $1 = |\{x \in \mathbb{T} \mid C\mu_N(x) > \log N\}| \leq \frac{A}{\log N} \rightarrow 0$ — **contradiction!**

9.4 The Banach principle

THEOREM. Let (T_n) be a sequence of bounded linear operators on $L^1(\mathbb{T})$, translation-invariant. Define

$$Mf(x) = \sup_{n \geq 1} |T_n f(x)|, \quad \|Mf\|_\infty < +\infty \quad \forall f \text{ trigonometric polynomial.}$$

If for every f ,

$$|\{x \mid Mf(x) < \infty\}| > 0, \quad (*)$$

then there exists A with

$$|\{x \mid Mf(x) > \lambda\}| \leq \frac{A}{\lambda} \|f\|_1.$$

SCHEME. Banach principle (together with $(*)$) $\Rightarrow \{f \mid \text{pointwise convergence holds}\}$ is a closed subset of $L^1 \Rightarrow \text{everything converges} \Rightarrow (\text{via the Corollary} + \text{Lemma above}) \text{ contradiction.}$

LEMMA. Let E_n be a sequence of measurable subsets of $\mathbb{T}$ with $\sum_{n \geq 1} |E_n| = +\infty$. Then there exists a sequence of points $(x_n)_{n \geq 1}$ such that

$$\sum_{n \geq 1} \mathbf{1}_{E_n}(x + x_n) = \infty \quad \text{a.e.}$$

9.4.1 Proof

By contradiction, assume there is a sequence $(f_j)_{j \geq 1} \subseteq L^1(\mathbb{T})$ with $\|f_j\|_{L^1} = 1$ such that for each j there is λ_j with

$$|\{x \mid Mf_j > \lambda_j\}| = |E_j| > \frac{2^j}{\lambda_j}.$$

By monotone convergence and the definition of M ,

$$\lim_{m \rightarrow +\infty} \left| \left\{ x \mid \sup_{1 \leq n \leq m} |T_n f_j(x)| > \lambda_j \right\} \right| > \frac{2^j}{\lambda_j}.$$

Hence for each j there is M_j with

$$\|T\| \geq \left| \left\{ x \mid \sup_{1 \leq n \leq M_j} |T_n f_j(x)| > \lambda_j \right\} \right| > \frac{2^j}{\lambda_j},$$

and WLOG f_j is a trigonometric polynomial, by continuity of the T_k .

(Side note: $\sigma_N f = K_N * f$, with the Fejér kernel $K_N = \frac{1}{N} \sum_{n=0}^{N-1} D_n \cdot$)

Choose integers $m_j \geq 1$ with

$$m_j \leq \frac{\lambda_j}{2^j} \leq m_j + 1 \implies \sum_{j=1}^{\infty} m_j |E_j| = \infty.$$

By the Lemma there exist points $x_{j,\ell}$, $1 \leq \ell \leq m_j$, with

$$\sum_{j=1}^{\infty} \sum_{\ell=1}^{m_j} \mathbf{1}_{E_j}(x - x_{j,\ell}) = \infty \quad \text{for a.e. } x \in \mathbb{T}.$$

Set $\delta_j = \frac{1}{j^2 m_j}$ and define the random function

$$f(x) = \sum_{j=1}^{\infty} \sum_{\ell=1}^{m_j} \pm \delta_j f_j(x - x_{j,\ell}).$$

The translations move all the bad sets together, so the construction is stable under translation; the random signs ensure that the terms do not cancel one another.

Then

$$\|f\|_{L^1} \leq \sum_{j=1}^{\infty} m_j \delta_j = \sum_j \frac{1}{j^2} < +\infty.$$

Let $x \in \mathbb{T}$ be such that $x \in E_j + x_{j,\ell}$ for infinitely many j , with $\ell = \ell(j)$; let J be the set of all such j .

Since T_n is translation invariant,

$$M[f_j(x - x_{j,\ell})] > \lambda_j \quad \text{for } x \in E_j + x_{j,\ell},$$

so there exists $n = n(j, x)$ with

$$|T_{n(j,x)} f_j(x - x_{j,\ell})| > \lambda_j \quad \text{on } E_j + x_{j,\ell}.$$

But

$$T_n f(x) = \sum_{j=1}^{\infty} \sum_{\ell=1}^{m_j} \pm \delta_j T_n f_j(x - x_{j,\ell}),$$

so

$$\begin{aligned} T_{n(j,x)} f(x) &= \sum_{k=1}^{\infty} \sum_{\ell=1}^{m_k} \pm \delta_k T_{n(j,x)} f_k(x - x_{k,\ell}) \\ &= \underbrace{\pm \delta_j T_{n(j,x)} f_j(x - x_{j,\ell})}_{> \delta_j \lambda_j \text{ in modulus}} + \underbrace{\text{other terms}}_{\text{expectation 0 with respect to the signs}}. \end{aligned}$$

Therefore

$$\mathbb{P}(|T_{n(j,x)} f(x)| > \delta_j \lambda_j) \geq \frac{1}{2} \implies \mathbb{P}(Mf = +\infty) \geq \frac{1}{2}.$$

Geometrically, half of the mass lies beyond $-\delta_j \lambda_j$, while the other half lies beyond $+\delta_j \lambda_j$.

9.4.2 The divergence event has full probability

PROOF. We show $\{Mf = +\infty\} \in \mathcal{F}_{\infty}$ (a tail event). Write

$$\begin{aligned} Mf(x) &= M \left(\sum_{k=1}^{\infty} \sum_{\ell=1}^{m_k} \pm \delta_k f_k(x - x_{k,\ell}) \right) \\ &= M \left(\underbrace{\sum_{k \geq N} \sum_{\ell=1}^{m_k} \pm \delta_k f_k(x - x_{k,\ell})}_{\text{tail}} \right) + M(\text{remainder}). \end{aligned}$$

where the remainder is a **finite** sum. Then:

- $Mf(x) = \infty \implies M(\text{tail}) = \infty$;
- $M(\text{tail}) = \infty \implies M(\text{tail}) \leq Mf(x) + M(\text{finite terms})$, so $Mf(x) = \infty$.

Hence $\{Mf(x) = \infty\} = \{M(\text{tail}) = \infty\}$, which belongs to $\mathcal{F}_{\infty}$. By the zero-one law and the bound $\geq \frac{1}{2}$ above,

$$\mathbb{P}(Mf(x) = \infty) = 1 \quad \text{for every } x.$$

Consequently

$$\begin{aligned} \int_0^1 \mathbb{P}(Mf(x) = +\infty) dx &= 1 \\ &= \int_0^1 \int_{\Omega} \mathbf{1}_{\{Mf(x)=\infty\}} d\omega dx \\ &= \int_{\Omega} \int_0^1 \mathbf{1}_{\{Mf(x)=\infty\}} dx d\omega \\ &= \mathbb{E}(|\{Mf(x) = \infty\}|). \end{aligned}$$

so $|\{Mf(x) = \infty\}| = 1$ for a.e. choice of signs ± 1 . $\square$

9.5 A measure-theoretic corollary and the construction

COROLLARY. If $\{S_N f\}$ converges for all $f \in L^1$, then there exists A' such that

$$|\{x \in \mathbb{T} \mid C\mu(x) > \lambda\}| \leq \frac{A'}{\lambda} \|\mu\|.$$

PROOF. We start from

$$|\{x \mid Cf(x) > \lambda\}| \leq \frac{A}{\lambda} \|f\|_{L^1(\mathbb{T})}.$$

Consider $f = V_N * \mu$, where

$$V_N(x) = K_N(x) + K_N(x)e(Nx) + K_N(x)e(-Nx)$$

(the de la Vallée Poussin type kernel). Since $\|K_N * \mu\|_{L^1} \leq \|\mu\|$, we get

$$\|V_N * \mu\|_{L^1} \leq 3 \|\mu\|_{L^1} \implies |\{x : |C(V_N * \mu)| > \lambda\}| \leq \frac{3A}{\lambda} \|\mu\|_M.$$

Hence

$$\lim_{M \rightarrow \infty} \left| \left\{ x \mid \sup_{1 \leq n \leq M} |S_n(V_N * \mu)(x)| > \lambda \right\} \right| \leq \frac{3A}{\lambda} \|\mu\|_M,$$

and therefore

$$\left| \left\{ x \mid \sup_{1 \leq n \leq M} \dots \right\} \right| \leq \frac{3A}{\lambda} \|\mu\|_M,$$

letting $n = M$, $n \rightarrow +\infty$. $\square$

9.5.1 A first construction

Set

$$\mu_N = \frac{1}{N} \sum_{j=1}^N \delta_{x_{j,N}}, \quad \left| x_{j,N} - \frac{j}{N} \right| < \frac{1}{N^2},$$

i.e. the points $x_{j,N}$ are tiny perturbations of the equispaced points $0, \frac{1}{N}, \frac{2}{N}, \dots, 1$.

Then

$$\begin{aligned} S_n \mu_N(x) &= D_n * \mu_N(x) = \int D_n(x-y) d\mu_N(y) \\ &= \frac{1}{N} \sum_{j=1}^N D_n(x - x_{j,N}) \\ &= \frac{1}{N} \sum_{j=1}^N \frac{\sin((2n+1)\pi(x - x_{j,N}))}{\sin(\pi(x - x_{j,N}))}. \end{aligned}$$

Choosing x_j so that $|\sin(\cdot)| \geq \frac{1}{2}$ in the numerator gives

$$\begin{aligned} |S_n \mu_N(x)| &\geq \frac{1}{2N} \sum_{j=1}^N \frac{1}{|\sin(\pi(x - x_{j,N}))|} \\ &\geq \frac{c}{N} \sum_{j=1}^N \frac{1}{j/N} = c \sum_{j=1}^N \frac{1}{j} \gtrsim c \log N. \end{aligned}$$

Remark 9.1. The estimate involving $\sin(\pi(x - x_{j,N}))$ requires the points $x_{j,N}$ to be ordered relative to x ; this is the geometric input behind the bound by j/N .

9.6 A refined almost-everywhere divergence construction

Write

$$Cf(x) = \sup_{n \geq 0} |S_n f(x)|$$

for the maximal partial-sum operator.

COROLLARY. If $\{S_N f\}$ converges a.e. (in fact: *on a set of positive measure*) for all f , then

$$|\{x \in \mathbb{T} \mid Cf(x) > \lambda\}| \leq \frac{A}{\lambda} \|f\|_{L^1} \quad (****)$$

and the same with f replaced by μ and $\|f\|_{L^1}$ by $\|\mu\|_{\mathbb{T}}$, for $\mu \in M(\mathbb{T})$.

To show that there exists $f \in L^1$ whose Fourier series diverges a.e. (on a set of positive measure), it suffices to show that (****) **fails**.

EQUIDISTRIBUTION LEMMA. Assume $(\theta_1, \dots, \theta_d) \in \mathbb{T}^d$ is incommensurate; that is, for every $(n_1, \dots, n_d) \in \mathbb{Z}^d \setminus \{0\}$,

$$n_1 \theta_1 + \dots + n_d \theta_d \notin \mathbb{Z}.$$

Then the orbit $\{(n\theta_1, \dots, n\theta_d) \bmod \mathbb{Z}^d\}$ is dense in $\mathbb{T}^d$.

CONSTRUCTION. There exists $c > 0$ such that for every N there is a probability measure μ_N on $\mathbb{T}$ with

$$\limsup_n \frac{|S_n \mu_N(x)|}{\log N} > c \quad \text{a.e. } x.$$

If so, and if (****) held, then with $\lambda_N \sim \log N$,

$$1 = |\{x \mid C\mu_N(x) \geq \lambda_N\}| \leq \frac{A}{\lambda_N} \|\mu_N\| = \frac{A}{\lambda_N} \longrightarrow 0,$$

a contradiction. $\square$

PROOF OF THE CONSTRUCTION. Set

$$\mu_N = \frac{1}{N} \sum_{j=1}^N \delta_{x_{j,N}}, \quad \left| x_{j,N} - \frac{j}{N} \right| \leq \frac{1}{N^2}, \quad j = 1, \dots, N.$$

Choose $(x_{j,N})_{j=1}^N$ to be **incommensurate** (this is possible since the set of commensurate vectors — those satisfying $n_1 x_1 + \dots + n_d x_d \in \mathbb{Z}$ — has measure zero).

Then choose n so that $(n(x - x_{j,N}))_{j=1}^N$ is also incommensurate (for a.e. x). Hence

$$((2n+1)(x - x_{j,N}))_{j=1}^N \bmod \mathbb{Z}^d \text{ is dense in } \mathbb{T}^N,$$

which implies

$$\sin((2n+1)\pi(x - x_{j,N})) \geq \frac{1}{2} \quad \text{for infinitely many } n, \text{ a.e. } x.$$

Therefore

$$\begin{aligned}
|S_n \mu_N(x)| &= |D_n * \mu_N(x)| \\
&= \left| \int_{\mathbb{T}} \frac{\sin((2n+1)\pi(x-y))}{\sin(\pi(x-y))} d\mu_N(y) \right| \\
&= \left| \frac{1}{N} \sum_j \frac{\sin((2n+1)\pi(x-x_{j,N}))}{\sin(\pi(x-x_j))} \right|. \\
&\geq \frac{1}{2N} \sum_{j=1}^N \frac{1}{|\sin(\pi(x-x_{j,N}))|} \geq \frac{c}{N} \sum_j \frac{1}{|x-x_{j,N}|} \geq \frac{c'}{N} \sum_j \frac{1}{j/N} \geq c \log N. \quad \square
\end{aligned}$$

10 Poisson Integrals and Boundary Values

10.1 The Hilbert transform and the Poisson kernel

On $\mathbb{R}$, the Hilbert transform is

$$Hf(x) = \text{p. v.} \frac{1}{\pi} \int_{\mathbb{R}} \frac{f(y)}{x-y} dy.$$

On $\mathbb{T}$, its periodic analogue is

$$Hf(x) = \text{p. v.} \frac{1}{\pi} \int_{-1/2}^{1/2} \cot\left(\frac{x-y}{2}\right) f(y) dy.$$

On the Fourier side, both are described by the same sign-type multiplier, continuous on $\mathbb{R}$ and sampled at the integer frequencies on $\mathbb{T}$.

Its connection with analytic functions can be formulated as follows. Given a real-valued $f \in L^p(\mathbb{T})$, one seeks an analytic function $F : \mathbb{D} \rightarrow \mathbb{C}$ satisfying

$$F|_{\partial\mathbb{D}} = f + ig \quad (f = \text{real part})?$$

Such a function is obtained from the Schwarz integral

$$F(z) = \int_{-1/2}^{1/2} \frac{e^{2\pi i x} + z}{e^{2\pi i x} - z} f(x) dx$$

1. By dominated convergence, F is analytic.
2. $F(0) = \int_{-1/2}^{1/2} f(x) dx = \widehat{f}(0) \in \mathbb{R}$.

Splitting into real and imaginary parts,

$$F(z) = \int_{-1/2}^{1/2} \text{Re}\left(\frac{e^{2\pi i x} + z}{e^{2\pi i x} - z}\right) f(x) dx + i \int_{-1/2}^{1/2} \text{Im}\left(\frac{e^{2\pi i x} + z}{e^{2\pi i x} - z}\right) f(x) dx.$$

With $z = re^{2\pi iy}$:

$$\begin{aligned} \frac{e^{2\pi ix} + z}{e^{2\pi ix} - z} &= \frac{e^{2\pi i(x-y)} + r}{e^{2\pi i(x-y)} - r} \\ &= \frac{(e^{2\pi i(x-y)} + r)(e^{-2\pi i(x-y)} - r)}{|e^{2\pi i(x-y)} - r|^2} \\ &= \underbrace{\frac{1 - r^2}{|e^{2\pi i(x-y)} - r|^2}}_{P_r(x-y)} - \underbrace{\frac{2ir \sin(2\pi(x-y))}{|e^{2\pi i(x-y)} - r|^2}}_{iQ_r(x-y)}. \end{aligned}$$

Hence

$$\operatorname{Re} F(re^{2\pi iy}) = P_r * f(y),$$

where $P_r(t) = P_r(-t)$, peaks at $t = 0$, and becomes singular as $r \rightarrow 1$.

PROPOSITION. P_r is an approximate identity, $0 \leq r < 1$.

PROOF. $P_r(t) \geq 0$.

$$\text{Claim: } P_r(t) = \sum_{k \in \mathbb{Z}} r^{|k|} e^{2\pi ikt}.$$

Proof of claim:

$$\begin{aligned} \sum_k r^{|k|} e^{2\pi ikt} &= 1 + \sum_{k \geq 1} r^k e^{2\pi ikt} + \sum_{k=1}^{\infty} r^k e^{-2\pi ikt} \\ &= 1 + \frac{re^{2\pi it}}{1 - re^{2\pi it}} + \frac{re^{-2\pi it}}{1 - re^{-2\pi it}} = P_r(t). \end{aligned}$$

Moreover,

$$\int_{\mathbb{T}} P_r(t) dt = \widehat{P_r}(0) = 1 = \|P_r\|_{L^1},$$

and for all $\delta > 0$,

$$\sup_{|\theta| > \delta} |P_r(t)| \rightarrow 0 \quad \text{as } r \rightarrow 1.$$

Indeed $P_r(t)$ is decreasing as a function of $|t|$, so

$$\sup_{|\theta| > \delta} |P_r(t)| = P_r(\delta) = \frac{1 - r^2}{|r - e^{2\pi i\delta}|^2} \xrightarrow{r \rightarrow 1} 0. \quad \square$$

Consequently

$$\operatorname{Re} F(re^{2\pi iy}) = P_r * f \longrightarrow f \quad \text{in } L^p.$$

10.2 Almost-everywhere convergence of Poisson integrals

CLAIM. $P_r * f \xrightarrow{r \rightarrow 1} f$ for a.e. $y \in \mathbb{T}$.

SKETCH. It suffices to show the weak-type bound

$$\left| \left\{ \sup_{r < 1} |P_r * f| > \lambda \right\} \right| \leq \frac{A}{\lambda} \|f\|_{L^1}.$$

The key pointwise domination is

$$|K * f(x)| \leq \|K\|_{L^1} Mf(x) \quad (M = \text{Hardy-Littlewood maximal function})$$

valid whenever K is even and decreasing in t (a "bump" peaked at the origin).

Baby case: $K = \sum_i a_i \mathbf{1}_{I_i}$ with all intervals I_i centered at 0. Then

$$|K * f| \leq \sum_i a_i (\mathbf{1}_{I_i} * f) \leq \sum_i a_i |I_i| Mf \leq \|K\|_{L^1} Mf.$$

General case: approximate by the baby case. $\square$

IMAGINARY PART. Similarly,

$$\operatorname{Im} F(re^{2\pi i y}) = Q_r * f \longrightarrow Hf, \quad Q_r(y) = \frac{2r \sin(2\pi y)}{|1 - re^{2\pi i y}|^2} \xrightarrow{r \rightarrow 1} \cot\left(\frac{y}{2}\right).$$

Part IV

Calderón–Zygmund Theory and BMO

11 Homogeneous Singular Integrals

11.1 Dini continuity and the Hörmander condition

Setting: $K(x) = \frac{\Omega(x/|x|)}{|x|^d}$ with Ω defined on S^{d-1} , $\int_{S^{d-1}} \Omega d\sigma = 0$, and Ω having modulus of continuity ω .

GOAL (III)' $\Rightarrow$ (III): the Dini condition

$$\int_0^1 \frac{\omega(\delta)}{\delta} d\delta < \infty$$

implies the Hörmander condition

$$\int_{|x| \geq 2|y|} |K(x-y) - K(x)| dx \leq B < \infty.$$

11.1.1 Splitting the difference

$$K(x-y) - K(x) = \frac{\Omega\left(\frac{x-y}{|x-y|}\right) - \Omega\left(\frac{x}{|x|}\right)}{|x-y|^d} + \Omega\left(\frac{x}{|x|}\right) \left[\frac{1}{|x-y|^d} - \frac{1}{|x|^d} \right].$$

The first term measures the **angular** variation of Ω , the second the **radial** variation of $|x|^{-d}$. Handle them separately.

11.1.2 Radial part

Claim: $\int_{|x| \geq 2|y|} \left| \frac{1}{|x-y|^d} - \frac{1}{|x|^d} \right| dx \leq C.$

Put $F(t) = \frac{1}{|x-ty|^d}$, $t \in [0, 1]$. By the fundamental theorem of calculus,

$$\left| \frac{1}{|x-y|^d} - \frac{1}{|x|^d} \right| = |F(1) - F(0)| \leq \int_0^1 |F'(t)| dt \leq d|y| \sup_{t \in [0, 1]} \frac{1}{|x-ty|^{d+1}}.$$

On the region $|x| \geq 2|y|$ we have $|x-ty| \geq |x| - |y| \geq \frac{|x|}{2}$, hence

$$\int_{|x| \geq 2|y|} \left| \frac{1}{|x-y|^d} - \frac{1}{|x|^d} \right| dx \leq d 2^{d+1} |y| \int_{|x| \geq 2|y|} |x|^{-d-1} dx \leq C,$$

since $\int_{|x| \geq R} |x|^{-d-1} dx \simeq R^{-1}$ and $R = 2|y|$. Note this bound is **independent of y** — this is the scale invariance of the estimate.

11.1.3 Angular part

First, the unit vectors are close when $|x| \geq 2|y|$:

$$\begin{aligned} \left| \frac{x-y}{|x-y|} - \frac{x}{|x|} \right| &= \frac{||x|(x-y) - |x-y|x||}{|x||x-y|} \\ &\leq \frac{|x||y| + |x||x| - |x-y||}{|x||x-y|} \\ &\leq \frac{2|x||y|}{|x||x-y|} \leq C \frac{|y|}{|x|}. \end{aligned}$$

where we used $||x| - |x-y|| \leq |y|$ and $|x-y| \geq |x|/2$.

Therefore $|\Omega(\frac{x-y}{|x-y|}) - \Omega(\frac{x}{|x|})| \leq \omega\left(C \frac{|y|}{|x|}\right)$, and

$$\int_{|x| \geq 2|y|} \frac{|\Omega(\frac{x-y}{|x-y|}) - \Omega(\frac{x}{|x|})|}{|x-y|^d} dx \leq C' \int_{|x| \geq 2|y|} \frac{\omega\left(C \frac{|y|}{|x|}\right)}{|x|^d} dx \leq C'' \int_0^{C/2} \frac{\omega(\delta)}{\delta} d\delta < +\infty.$$

(The last step: polar coordinates $dx = r^{d-1} dr d\sigma$ turns the integral into $c \int_{2|y|}^{\infty} \omega(C|y|/r) \frac{dr}{r}$; substituting $\delta = C|y|/r$ gives $\frac{dr}{r} = -\frac{d\delta}{\delta}$ and the range $0 < \delta \leq C/2$.)

□

11.2 Translation invariance and dilation structure

PROP. Let $T : L^2 \rightarrow L^2$ be bounded and linear, commuting with all translations,

$$T \circ \tau_x = \tau_x \circ T, \quad (\tau_x f)(y) = f(x+y).$$

Then there exists $m \in L^\infty$ such that $\widehat{Tf} = m \widehat{f}$.

11.2.1 Homogeneity of the multiplier

$K(x) = \frac{\Omega(x)}{|x|^d}$ is homogeneous of degree $-d$: $K(\varepsilon x) = \varepsilon^{-d} K(x)$.

For $\varepsilon > 0$ define the **unitary** dilation on $L^2(\mathbb{R}^d)$:

$$(D_\varepsilon f)(x) = \varepsilon^{-d/2} f\left(\frac{x}{\varepsilon}\right).$$

Let $Tf = \text{p.v. } K * f$. Then $T \circ D_\varepsilon = D_\varepsilon \circ T$:

$$\begin{aligned}
D_\varepsilon(Tf)(x) &= \varepsilon^{-d/2} T f\left(\frac{x}{\varepsilon}\right) = \varepsilon^{-d/2} \text{p.v.} \int K\left(\frac{x}{\varepsilon} - y\right) f(y) dy \\
&= \varepsilon^{d/2} \text{p.v.} \int K(x - \varepsilon y) f(y) dy & \left[K\left(\frac{x - \varepsilon y}{\varepsilon}\right) = \varepsilon^d K(x - \varepsilon y) \right] \\
&= \text{p.v.} \int K(x - z) \varepsilon^{-d/2} f\left(\frac{z}{\varepsilon}\right) dz & [z = \varepsilon y] \\
&= T D_\varepsilon f(x).
\end{aligned}$$

On the Fourier side this commutation says the multiplier m satisfies $m(\lambda\xi) = m(\xi)$, i.e. m is homogeneous of degree 0.

11.2.2 Truncated kernels

For $0 < \varepsilon < \eta < \infty$ set

$$K_{\varepsilon,\eta}(x) = \begin{cases} \Omega(x)/|x|^d, & \varepsilon \leq |x| \leq \eta, \\ 0, & \text{else.} \end{cases}$$

These are integrable, so $\widehat{K_{\varepsilon,\eta} * f} = \widehat{K_{\varepsilon,\eta}} \widehat{f}$ makes sense. The two things to prove are:

1. (i) $|\widehat{K_{\varepsilon,\eta}}| \leq A$ uniformly in ε, η ;
2. (ii) for $x \neq 0$, $\lim_{\varepsilon \rightarrow 0, \eta \rightarrow \infty} \widehat{K_{\varepsilon,\eta}}(x) = m(x)$.

11.3 Fourier multipliers of truncated kernels

Write $x = Rx'$ with $x' = x/|x|$, $R = |x|$, and $y = ry'$ with $y' = y/|y|$.

Define

$$I_{\varepsilon,\eta}(x, y') = \int_\varepsilon^\eta \frac{e^{2\pi i R r x' \cdot y'} - \cos(2\pi R r)}{r} dr.$$

The subtraction of $\cos(2\pi R r)$ is legitimate because $\int_{S^{d-1}} \Omega(y') d\sigma(y') = 0$ — it costs nothing and it makes the integral converge at $r \rightarrow 0$.

IMAGINARY PART.

$$\int_\varepsilon^\eta \frac{\sin(2\pi R r x' \cdot y')}{r} dr$$

is bounded **uniformly** in ε, η, R (Dirichlet integral), and converges to

$$\int_0^\infty \frac{\sin t}{t} dt \cdot \text{sgn}(x' \cdot y') = \frac{\pi}{2} \text{sgn}(x' \cdot y').$$

REAL PART.

$$\int_\varepsilon^\eta \frac{\cos(2\pi R r x' \cdot y') - \cos(2\pi R r)}{r} dr$$

is bounded in absolute value by $C \log \frac{1}{|x' \cdot y'|} + C$ for some constant C , and converges to $\log \frac{1}{|x' \cdot y'|}$.

ASSEMBLING. Since $\int_{S^{d-1}} \Omega d\sigma = 0$,

$$\widehat{K_{\varepsilon,\eta}}(x) = \int_{S^{d-1}} \int_\varepsilon^\eta \frac{e^{2\pi i R r x' \cdot y'}}{r} \Omega(y') dr d\sigma(y') = \int_{S^{d-1}} I_{\varepsilon,\eta}(x, y') \Omega(y') d\sigma(y').$$

Hence the uniform bound

$$|\widehat{K_{\varepsilon,\eta}}(x)| \leq \int_{S^{d-1}} |I_{\varepsilon,\eta}(x, y')| |\Omega(y')| d\sigma(y') \leq A \int_{S^{d-1}} \left[1 + \log \frac{1}{|x' \cdot y'|} \right] |\Omega(y')| d\sigma(y') \leq A',$$

the last integral being finite because $\log \frac{1}{|x' \cdot y'|}$ is integrable on the sphere.

Consequently $\widehat{K_{\varepsilon,\eta}}(x) \rightarrow m(x)$ pointwise for $x \neq 0$, and for $f \in L^2(\mathbb{R}^d)$:

$$K_{\varepsilon,\eta} * f \text{ converges in } L^2, \text{ with Fourier transform } \rightarrow m\widehat{f}.$$

(Order of limits: first $\eta \rightarrow +\infty$ with ε fixed, then $\varepsilon \rightarrow 0$.)

12 Maximal Singular Integrals

DEF. $T^*f(x) = \sup_{\varepsilon > 0} |T_\varepsilon f(x)|$, where $T_\varepsilon f(x) = \int_{|x-y| > \varepsilon} K(x-y)f(y) dy$.

THEOREM. Let Ω be as before.

- (a) If $1 \leq p < \infty$, then $\lim_{\varepsilon \rightarrow 0} T_\varepsilon f(x)$ exists for a.e. x , for every $f \in L^p$.
- (b) $T^* : L^p(\mathbb{R}^d) \rightarrow L^p(\mathbb{R}^d)$ is bounded for $1 < p < \infty$.
- (c) $|\{x \in \mathbb{R}^d : |T^*f(x)| > \lambda\}| \leq \frac{C}{\lambda} \|f\|_{L^1(\mathbb{R}^d)}$ for $f \in L^1$, $\lambda > 0$.

(a) follows from (c) by the standard density + maximal-inequality argument.

12.1 Calderón–Zygmund decomposition and the good part

Let $f \in L^1$, $\lambda > 0$. Perform the **Calderón–Zygmund decomposition at height λ** :

$$f = g + b,$$

- $|g| \leq \lambda$ (the good part),
- b supported in the bad cubes $\{Q_j\}$, with $\int_{Q_j} b = 0$ for all j ,
- $\sum_j |Q_j| \leq C \frac{\|f\|_{L^1}}{\lambda}$.

DEF. $Q_j^* = 2Q_j$ (concentric dilate), and $F^* = [\bigcup_j Q_j^*]^c$.

Split:

$$|\{x : T^*f > \lambda\}| \leq |\{x : T^*b > \frac{\lambda}{2}\}| + |\{x : T^*g > \frac{\lambda}{2}\}|.$$

The **good part** is handled by the L^2 theory (Chebyshev + L^2 boundedness of T^*):

$$|\{x : T^*g > \frac{\lambda}{2}\}| \leq C \frac{\|g\|_{L^2}^2}{\lambda^2} \leq C \frac{1}{\lambda} \|g\|_{L^1} \leq C \frac{1}{\lambda} \|f\|_{L^1},$$

using $\|g\|_{L^2}^2 \leq \|g\|_\infty \|g\|_1 \leq \lambda \|g\|_1$ and $\|g\|_1 \leq C \|f\|_1$.

It therefore remains to prove

$$|\{x : T^*b > \lambda\}| \leq C \frac{\|f\|_{L^1}}{\lambda}.$$

12.2 Cancellation estimate for the bad part

CLAIM. For all $x \in F^*$,

$$|T^*b(x)| \leq \underbrace{\sum_j \int_{Q_j} |K(x-y) - K(x-y_j)| |b(y)| dy}_{=: \Sigma(x)} + C Mf(x),$$

where y_j is the **center** of Q_j .

12.3 Consequences of the cancellation estimate

$$|\{T^*b \geq \frac{\lambda}{2}\}| \leq \underbrace{|\{x \in F^* : \Sigma(x) > \frac{\lambda}{4}\}|}_{\textcircled{1}} + \underbrace{|\{x \in F^* : Mf(x) > \frac{\lambda}{4C}\}|}_{\leq C\|f\|_1/\lambda \text{ (weak (1,1) of } M)}} + \underbrace{|(F^*)^c|}_{\leq C\|f\|_1/\lambda}$$

The third term is controlled because $|(F^*)^c| \leq \sum_j |Q_j^*| = 2^d \sum_j |Q_j| \leq C\|f\|_1/\lambda$.

For $\textcircled{1}$, use Chebyshev and Fubini:

$$\begin{aligned} \textcircled{1} &\leq \frac{C}{\lambda} \int_{F^*} |\Sigma(x)| dx = \frac{C}{\lambda} \sum_j \int_{Q_j} \underbrace{\int_{F^*} |K(x-y) - K(x-y_j)| dx}_{\leq B \text{ (Hörmander)}} |b(y)| dy \\ &\leq \frac{CB}{\lambda} \sum_j \int_{Q_j} |b(y)| dy \leq \frac{CB}{\lambda} \|f\|_{L^1}. \end{aligned}$$

The inner integral is bounded by the Hörmander constant B precisely because $x \in F^*$ means $x \notin 2Q_j$, so $|x - y_j| \geq 2|y - y_j|$ for $y \in Q_j$. $\square$

12.4 Geometric proof of the cancellation estimate

Fix $x \in F^*$ and look at

$$T_\varepsilon b(x) = \sum_j \int_{Q_j} K_\varepsilon(x-y) b(y) dy, \quad K_\varepsilon(z) = K(z) \mathbf{1}_{\{|z| > \varepsilon\}}.$$

Classify each cube Q_j by its position relative to the ball $B(x, \varepsilon)$:

CASE 1 — $Q_j \subseteq B(x, \varepsilon)$. For every $y \in Q_j$ we have $|x - y| < \varepsilon$, so $K_\varepsilon(x - y) = 0$ and

$$\int_{Q_j} K_\varepsilon(x - y) b(y) dy = 0.$$

Such cubes contribute nothing.

CASE 2 — $Q_j \cap B(x, \varepsilon) = \emptyset$. Here $K_\varepsilon(x - y) = K(x - y)$ on all of Q_j , so we may use the cancellation $\int_{Q_j} b = 0$:

$$\begin{aligned} \left| \int_{Q_j} K_\varepsilon(x - y) b(y) dy \right| &= \left| \int_{Q_j} K(x - y) b(y) dy \right| \\ &= \left| \int_{Q_j} [K(x - y) - K(x - y_j)] b(y) dy \right| \\ &\leq \int_{Q_j} |K(x - y) - K(x - y_j)| |b(y)| dy. \end{aligned}$$

These are exactly the terms of $\Sigma(x)$.

CASE 3 — THE REMAINING CASE (Q_j MEETS BOTH $B(x, \varepsilon)$ AND ITS COMPLEMENT). Then the cube straddles the sphere $\partial B(x, \varepsilon)$, and since $x \in F^*$ (so x is outside $2Q_j$, hence Q_j is not too big compared to its distance from x), we get for all $y \in Q_j$

$$\frac{\varepsilon}{10} \leq \text{dist}(x, y) \leq 10\varepsilon.$$

Therefore

$$\int_{Q_j} |K_\varepsilon(x - y)b(y)| dy \leq C \int_{Q_j} \frac{|b(y)|}{|x - y|^d} dy \leq \frac{C}{\varepsilon^d} \int_{Q_j} |b(y)| dy \leq C \frac{1}{(10\varepsilon)^d} \int_{B(x, 10\varepsilon) \cap Q_j} |b(y)| dy,$$

which is $\leq C M b(x) \leq C M f(x)$ — an average over a ball centered at x , i.e. controlled by the **maximal function**. (Only boundedly many cubes fall into Case 3.)

Summing over all j and taking $\sup_{\varepsilon > 0}$ gives the claim. $\square$

13 BMO and Endpoint Estimates

DEF. For $f \in L^1_{loc}$, $f \in \text{BMO}$ if

$$\|f\|_{\text{BMO}} := \sup_Q \frac{1}{|Q|} \int_Q |f - f_Q| dx < \infty.$$

NOTE. $f \in L^\infty \Rightarrow f \in \text{BMO}$, with $\|f\|_{\text{BMO}} \leq 2\|f\|_\infty$.

BMO is a Banach space with norm $\|\cdot\|_{\text{BMO}}$ **modulo constant functions** (constants have BMO norm 0).

13.1 Averages over nested cubes grow only linearly

OBSERVATION. Let $Q_2 \subset Q_1$ be cubes with $\ell(Q_2) = \frac{1}{2}\ell(Q_1)$, $f \in \text{BMO}$. Then

$$|f_{Q_2} - f_{Q_1}| \leq \frac{1}{|Q_2|} \int_{Q_2} |f - f_{Q_1}| \leq \frac{|Q_1|}{|Q_2|} \cdot \frac{1}{|Q_1|} \int_{Q_1} |f - f_{Q_1}| \leq 2^d \|f\|_{\text{BMO}}.$$

Iterating: if $Q_n \subset Q_{n-1} \subset \cdots \subset Q_1 \subset Q_0$ with $\ell(Q_j) = \frac{1}{2}\ell(Q_{j-1})$, then

$$|f_{Q_n} - f_{Q_0}| \leq n 2^d \|f\|_{\text{BMO}}.$$

Thus geometric changes in scale affect the averages only arithmetically. This linear growth is precisely the source of the logarithmic blow-up allowed by BMO and, ultimately, of the John–Nirenberg inequality.

13.2 Equivalent norm with arbitrary constants

$$\|f\|_{\text{BMO}} = \sup_Q \inf_{c_Q} \frac{1}{|Q|} \int_Q |f - c_Q| dx.$$

NOTE. $\|f\|_{\text{BMO}} \leq \|f\|_{\text{BMO}} \leq 2\|f\|_{\text{BMO}}$.

Proof of the right inequality. For any constant c_Q ,

$$\frac{1}{|Q|} \int_Q |f - f_Q| \leq \frac{1}{|Q|} \int_Q |f - c_Q| + \frac{1}{|Q|} \int_Q |f_Q - c_Q| \leq \|f\|_{\text{BMO}} + |f_Q - c_Q| + \varepsilon,$$

and

$$|f_Q - c_Q| = \left| \frac{1}{|Q|} \int_Q (f - c_Q) \right| \leq \frac{1}{|Q|} \int_Q |f - c_Q|,$$

so taking the infimum over c_Q gives $\leq 2\|f\|_{\text{BMO}} + \varepsilon$. $\square$

The left inequality is immediate ($c_Q = f_Q$ is one admissible choice).

REMARK. Cubes $\rightsquigarrow$ balls: replacing cubes by balls gives an equivalent norm.

13.3 Where BMO sits

$$L^\infty \subsetneq \text{BMO} \subseteq \bigcap_{p \geq 1} L_{loc}^p$$

Related duality (Hardy spaces H^p):

- $(H^p)^* \simeq H^{p'}$,
- $(H^1)^* \simeq \text{BMO}$ (Fefferman duality),
- $\|f\|_{H^p} = \|f\|_{L^p} + \|Hf\|_{L^p}$.

13.4 Calderón–Zygmund operators map L^∞ to BMO

PROP. Let T be a Calderón–Zygmund operator. Then for all $f \in L^2 \cap L^\infty$,

$$\|Tf\|_{\text{BMO}} \leq C\|f\|_\infty, \quad \text{i.e.} \quad T : L^\infty \longrightarrow \text{BMO}.$$

Heuristically, a Calderón–Zygmund operator may send a bounded bump to a function with logarithmic growth. This explains why L^∞ is not preserved and why BMO is the natural endpoint space.

13.5 Local and far-field decomposition

Since $f \in L^2$, Tf is well-defined. Fix a ball $B_0 = B(x_0, R)$ and choose the constant

$$c_{B_0} = \int_{|x_0 - y| \geq 2R} K(x_0 - y) f(y) dy$$

(the "far field" evaluated at the center — this is the natural normalization, and by §7.2 we are free to use any constant).

Split $f = f \mathbf{1}_{\{|x_0 - y| \geq 2R\}} + f \mathbf{1}_{\{|x_0 - y| < 2R\}}$ (far part + near part). Write

$$\frac{1}{|B_0|} \int_{B_0} |Tf(x) - c_{B_0}| dx \leq I_{\text{far}} + I_{\text{near}},$$

where

$$I_{\text{far}} = \frac{1}{|B_0|} \int_{B_0} \left| \int_{|x_0-y| \geq 2R} (K(x-y) - K(x_0-y)) f(y) dy \right| dx,$$

$$I_{\text{near}} = \frac{1}{|B_0|} \int_{B_0} \left| T(f \mathbf{1}_{\{|x_0-y| < 2R\}})(x) \right| dx.$$

FIRST TERM (FAR FIELD — SMOOTHNESS/HÖRMANDER). For $x \in B_0$ and $|x_0 - y| \geq 2R$ we have $|x_0 - y| \geq 2|x - x_0|$, so the Hörmander condition applies:

$$I_{\text{far}} \leq \frac{\|f\|_{\infty}}{|B_0|} \int_{B_0} \int_{|x_0-y| \geq 2R} |K(x-y) - K(x_0-y)| dy dx$$

$$\leq CB \|f\|_{\infty}.$$

SECOND TERM (NEAR FIELD — HÖLDER + L^2 BOUNDEDNESS). By Cauchy–Schwarz and then $L^2 \rightarrow L^2$ boundedness of T :

$$I_{\text{near}} \leq \frac{1}{|B_0|^{1/2}} \left(\int_{B_0} |T(f \mathbf{1}_{B(x_0, 2R)})|^2 \right)^{1/2}$$

$$\leq \frac{C}{|B_0|^{1/2}} \left(\int |f \mathbf{1}_{B(x_0, 2R)}|^2 \right)^{1/2}$$

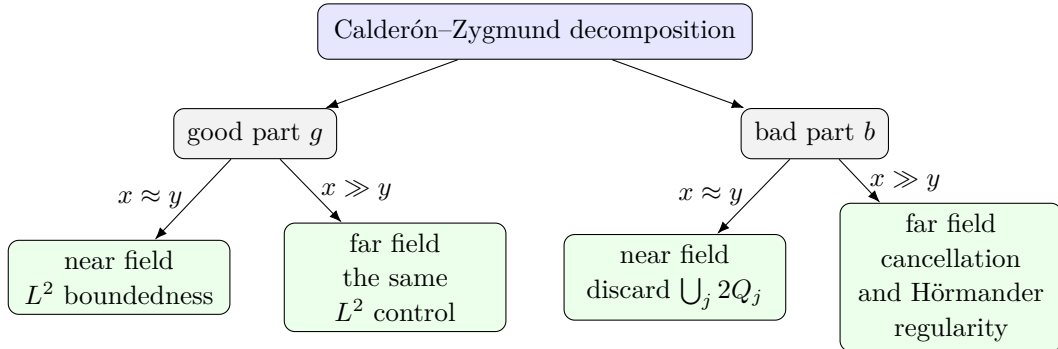
$$\leq C \|f\|_{\infty},$$

the last step because $|B(x_0, 2R)| \leq 2^d |B_0|$, so $\left(\int_{B(x_0, 2R)} |f|^2 \right)^{1/2} \leq \|f\|_{\infty} |B(x_0, 2R)|^{1/2} \leq C |B_0|^{1/2} \|f\|_{\infty}$.

Taking the supremum over all balls B_0 gives $\|Tf\|_{\text{BMO}} \leq C \|f\|_{\infty}$. $\square$

14 Calderón–Zygmund Proof Architecture

Every Calderón–Zygmund estimate decomposes along the same two-by-two grid: **which piece of f** (good/bad) $\times$ **which regime of the kernel** (x near y / x far from y).



- g , $x \approx y$ **and** $x \gg y$: the good part is bounded, so raw $L^2 \rightarrow L^2$ boundedness suffices; no cancellation needed.
- b , $x \approx y$: the singular part lives inside the bad cubes; one simply throws away $\bigcup 2Q_j$, whose measure is $\lesssim \|f\|_1 / \lambda$.

- $b, x \gg y$: away from the bad cubes, the mean-zero property of b plus the smoothness of K (Hörmander) gives the decay that makes the sum converge.

Part V

Multipliers and Littlewood–Paley Theory

15 Fourier Multipliers

15.1 The Mihlin multiplier theorem

A basic multiplier arises from the equation $\Delta u = f$. Consider the operator sending f to a second derivative of u :

$$T : f \mapsto \partial_i \partial_j u.$$

Taking Fourier transforms,

$$\widehat{Tf} = -(2\pi)^2 \xi_i \xi_j \widehat{u} = \frac{\xi_i \xi_j}{|\xi|^2} \widehat{f}(\xi).$$

So the associated multiplier is $m(\xi) = \frac{\xi_i \xi_j}{|\xi|^2}$, which is 0-homogeneous.

DEF. For $m : \mathbb{R}^d \rightarrow \mathbb{C}$ bounded, and $f \in \mathcal{S}(\mathbb{R}^d)$,

$$T_m f = (\widehat{f}(\xi) m(\xi))^\vee.$$

REM. T is an operator on $L^2(\mathbb{R}^d)$ with operator norm given by $\|m\|_\infty$.

DEF. We say $m : \mathbb{R}^d \rightarrow \mathbb{C}$ is a **Mihlin multiplier** if

$$\exists B > 0 \text{ s.t. } \forall \gamma \text{ multi-index with } |\gamma| \leq d+2, \quad \|\partial^\gamma m(\xi)\| \leq B |\xi|^{-|\gamma|}.$$

REM. If $m \in C^{d+2}(\mathbb{R}^d \setminus \{0\})$ is 0-homogeneous (i.e. $m(\lambda\xi) = m(\xi)$), then m is a Mihlin multiplier.

THEOREM 1. If m is a Mihlin multiplier, then $\exists C \sim_{d,p}$ such that for all $f \in \mathcal{S}(\mathbb{R}^d)$,

$$\|T_m f\|_p \leq CB \|f\|_p, \quad 1 < p < \infty.$$

REM. If m is a Mihlin multiplier, $\lambda > 0$, then the dilate $D_\lambda m(\xi) = m(\lambda\xi)$ is a Mihlin multiplier with the same constant B .

15.2 Reduction to Calderón–Zygmund kernels

DEF. A **Calderón–Zygmund (C–Z) kernel** is a function $K : \mathbb{R}^d \setminus \{0\} \rightarrow \mathbb{C}$ such that

1. $|K(x)| \leq B |x|^{-d}$;
2. $\int_{|x| \geq 2|y|} |K(x) - K(x-y)| dy \leq B$;
3. $\int_{r < |x| < R} K(x) dx = 0$.

THEOREM (C-Z).

$$Tf = \lim_{\varepsilon \rightarrow 0} \int_{|x-y| > \varepsilon} K(x-y) f(y) dy$$

is bounded on L^p with norm $\lesssim CB$.

LEMMA 3. If $K : \mathbb{R}^d \setminus \{0\} \rightarrow \mathbb{C}$ satisfies $|\nabla K| \leq B|x|^{-d-1}$, then condition (2) holds.

Proof.

$$\begin{aligned} \int_{|x| \geq 2|y|} |K(x) - K(x-y)| dy &\leq \int_{|x| \geq 2|y|} \underbrace{\int_0^1 |\nabla K(x+ty)| dt}_{|x+ty| \geq |x|/2 \geq |y|} |y| \\ &\leq \int_{|x| \geq 2|y|} B \left(\frac{|x|}{2} \right)^{-d-1} |y| \leq CB. \end{aligned}$$

Also note

$$T_m f = (m\hat{f})^\vee = m^\vee * f.$$

LEMMA 4. There exists $\psi \in C^\infty(\mathbb{R}^d)$ such that

1. $\psi \geq 0$, radial, with $\text{supp } \psi \subseteq \{\frac{1}{2} \leq |\xi| \leq 2\}$;
2. $\sum_{j \in \mathbb{Z}} \psi(2^{-j}\xi) = 1$ for $\xi \neq 0$.

Proof. Pick $\tilde{\psi}$ smooth, radial, with $\text{supp } \tilde{\psi} \subset \{\frac{1}{2} \leq |\xi| \leq 2\}$ and $\tilde{\psi} \geq 1$ on $\{\frac{2}{3} \leq |\xi| \leq \frac{4}{3}\}$. Let

$$\psi = \frac{\tilde{\psi}}{\sum_j \tilde{\psi}(2^{-j}\xi)}. \quad \square$$

LEMMA 5. Set

$$m_j(\xi) = m(\xi) \psi(2^{-j}\xi), \quad \text{so that } \sum_j m_j = m.$$

Define $K_j(x) = m_j^\vee(x)$ and $K(x) = \sum_{-N}^N K_j(x)$. Then (with constants **independent of K and N**):

1. $|K(x)| \leq CB|x|^{-d}$;
2. $|\nabla K(x)| \leq CB|x|^{-d-1}$.

15.3 Dyadic kernel estimates

CLAIM 1.

1. $|K_0(x)| \leq CB \min\{1, |x|^{-d-2}\}$;
2. $|\nabla K_0(x)| \leq CB \min\{1, |x|^{-d-2}\}$.

PROOF OF (I). It suffices to show

$$\|x_j^{d+2} K_0(x)\|_\infty \leq CB \quad \forall j, \quad \text{and} \quad \|K_0(x)\|_\infty \leq CB.$$

Now

$$\|x_j^{d+2} K_0(x)\|_\infty \leq \|(x_j^{d+2} K_0(x))^\wedge\|_{L^1} = C \|\partial_j^{d+2} m_0(\xi)\|_{L^1}.$$

Expanding the derivative of $m_0 = m \psi$:

$$\partial_j^{d+2}(m(\xi)\psi(\xi)) = \sum_{\ell=0}^{d+2} \binom{d+2}{\ell} \partial_j^\ell \psi \cdot \partial_j^{d+2-\ell} m,$$

and on $\text{supp } \psi$ (where $\frac{1}{2} \leq |\xi| \leq 2$),

$$|\partial_j^{d+2-\ell} m| \leq B |\xi|^{-(d+2-\ell)} \leq B \left(\frac{1}{2}\right)^{-(d+2-\ell)} \leq B 2^{d+2}.$$

Therefore

$$\|\partial_j^{d+2} m_0\|_{L^1} \leq C(d) B \|\mathbf{1}_{B(0,2)}\|_{L^1} \implies \text{(i)}.$$

PROOF OF (II). It suffices to show

$$\|x_j^{d+2} \partial_k K_0(x)\|_{L^\infty} \leq CB, \quad \|\partial_k K_0(x)\|_\infty \leq CB.$$

Then

$$\|x_j^{d+2} \partial_k K_0(x)\|_{L^\infty} \leq \|(x_j^{d+2} \partial_k K_0)^\wedge\|_{L^1} = C \|\partial_j^{d+2} [\xi_k m_0]\|_{L^1} \stackrel{\text{same argument}}{\leq} CB.$$

CLAIM 2.

1. $|K_j(x)| \leq CB 2^{jd} \min\{1, |2^j x|^{-d-2}\};$
2. $|\nabla K_j(x)| \leq CB 2^{j(d+1)} \min\{1, |2^j x|^{-d-2}\}.$

(The factor $2^{jd} / 2^{j(d+1)}$ comes from the change of variables $\xi \mapsto 2^{-j}\xi$; the exponent $-d-2$ comes from the $(d+2)$ -fold differentiation, as in Claim 1.)

PROOF OF THE LEMMA.

$$|K(x)| \leq \sum_{-N}^N |K_j(x)| \leq CB \sum_{j=-N}^N 2^{jd} \min\{1, |2^j x|^{-d-2}\}.$$

Split according to whether 2^j is below or above $|x|^{-1}$:

$$\begin{aligned} &\leq CB \sum_{2^j < |x|^{-1}} 2^{jd} + CB \sum_{2^j \geq |x|^{-1}} |2^j x|^{-d-2} 2^{jd} \\ &\leq CB |x|^{-d} + CB |x|^{-d-2} \sum_{2^j \geq |x|^{-1}} 2^{-2j} \leq CB |x|^{-d}. \quad \square \end{aligned}$$

(The ∇K estimate follows by the same splitting, using Claim 2(ii).)

15.4 Proof of the Mikhlin theorem

For $f \in \mathcal{S}(\mathbb{R}^d)$,

$$T_m f = \mathcal{F}^{-1}(m \hat{f}) = \mathcal{F}^{-1}\left(\sum_j m_j \hat{f}\right).$$

Writing $K^N = \sum_{-N}^N K_j$ (which is uniformly a C-Z kernel by Lemma 5),

$$\|T_m f\|_p = \|\lim f * K^N\|_p \leq \liminf \|f * K^N\|_p \leq CB \|f\|_p. \quad \blacksquare$$

16 Littlewood–Paley Theory

16.1 Dimension $d = 1$

For $f \in \mathcal{S}(\mathbb{R})$, define the **(rough) Littlewood–Paley projection**

$$\pi_j f(x) = \left(\mathbf{1}_{[2^j, 2^{j+1}] \cup (-2^{j+1}, -2^j]} \cdot \widehat{f} \right)^\vee, \quad j \in \mathbb{Z}.$$

Then (orthogonality on L^2):

$$\left\| \left(\sum_{j \in \mathbb{Z}} |\pi_j f|^2 \right)^{1/2} \right\|_{L^2(\mathbb{R})}^2 = \sum_{j \in \mathbb{Z}} \|\pi_j f\|_{L^2}^2 = \|f\|^2,$$

so heuristically $f \approx (\pi_j f)_{j \in \mathbb{Z}}$.

DEF. The rough Littlewood–Paley square function is

$$S_\pi f := \left(\sum_j |\pi_j f(x)|^2 \right)^{1/2}.$$

THEOREM. For $1 < p < +\infty$, there exists C_p such that

$$\frac{1}{C_p} \|f\|_p \leq \|S_\pi f\|_p \leq C_p \|f\|_p.$$

The mechanism behind the estimate is Khinchine's inequality:

$$\left\| \sum_{j=1}^N a_j r_j \right\|_p \approx \left(\sum a_j^2 \right)^{1/2}, \quad r_j = \text{Rademacher ("wave").}$$

Here $\pi_j f$ is a "wave" of frequency 2^j . Khinchine says: a sum of waves whose frequencies double at each step $\approx$ the square-sum of the coefficients:

$$|\pi_j f(x)| \longleftrightarrow a_j, \quad \pi_j f(x) \approx a_j e^{2\pi i 2^j x}.$$

Proof: not proven here. (One recovers π_j from **Hilbert transforms**: the sharp cutoff $\mathbf{1}_{[2^j, 2^{j+1}]}$ is a difference of shifted Hilbert-transform-type step multipliers — by shifting the Hilbert transform and subtracting, one recovers π_j .)

16.2 Higher dimensions

In higher dimensions, however, the sharp cutoff is no longer bounded.

FACT. $f \mapsto (\mathbf{1}_{B(0,1)} \widehat{f})^\vee$ is **unbounded** on L^p if $d \geq 2$ (the ball multiplier problem).

Instead, for each $j \in \mathbb{Z}$, define **smooth Littlewood–Paley projections**:

$$P_j f = (\psi_j \widehat{f})^\vee, \quad \psi \in C_c^\infty(\mathbb{R}^n \setminus \{0\}), \quad \sum_j \psi_j = \sum_j \psi(2^{-j} \xi) \equiv 1,$$

with $P_j f$ localized at frequency $\sim 2^j$. Define the smooth square function

$$S_P f = \left(\sum_j |P_j f|^2 \right)^{1/2}.$$

THEOREM. Let $1 < p < \infty$. There exists $C_{p,d}$ such that

$$\|S_P f\|_{L^p} \approx_{C_{p,d}} \|f\|_{L^p}.$$

Proof (uses Khinchine + Mikhlin).

Set $m(\xi) = \sum_{-N}^N r_j \psi_j(\xi)$.

CLAIM: $m(\xi)$ satisfies the Mikhlin multiplier condition. Indeed,

$$|D^r m(\xi)| \leq \sum_{-N}^N |D^r \psi_j(\xi)| = \sum_{-N}^N 2^{-j|r|} |D^r \psi(\xi/2^j)| \leq \sum_{-N}^N |\xi|^{-r} |D^r \psi(\xi/2^j)| \leq |\xi|^{-r} C_r,$$

using that for each ξ the supports of $\psi(\cdot/2^j)$ **overlap only ~ 3 times**.

UPPER BOUND. With $a_j = |P_j f(x)|$ and Khinchine,

$$\begin{aligned} \|S_P f\|_{L^p}^p &= \lim_{N \rightarrow \infty} \int \left(\sum_{-N}^N |P_j f|^2 \right)^{p/2} \leq C_p \int \mathbb{E} \left| \sum_j r_j P_j f(x) \right|^p dx \\ &= C_p \mathbb{E} \int \underbrace{\left| \sum_j r_j P_j f(x) \right|^p}_{=(m\widehat{f})^\vee} dx \leq C_{p,d} \|f\|_{L^p}, \end{aligned}$$

the last step by the Mikhlin theorem applied to m . This gives $\|S_P f\|_p \leq C_{p,d} \|f\|_p$.

LOWER BOUND (DUALITY). Find $\tilde{\psi}$ with $\tilde{\psi} \equiv 1$ on $\text{supp } \psi$; define $\tilde{\psi}_j$ analogously and

$$\tilde{P}_j g(x) = \mathcal{F}^{-1}(\tilde{\psi}_j \widehat{g})(x).$$

Then

$$\begin{aligned} \langle f, g \rangle &= \langle \mathcal{F}f, \mathcal{F}g \rangle = \int \widehat{f} \widehat{g} d\xi \\ &= \int \sum_j \psi_j \widehat{f} \widehat{g} d\xi \\ &= \sum_j \int \psi_j \widehat{f} \widetilde{\psi_j \widehat{g}} d\xi \\ &= \sum_j \langle \psi_j \widehat{f}, \widetilde{\psi_j \widehat{g}} \rangle = \sum_j \langle P_j f, \tilde{P}_j g \rangle. \end{aligned}$$

Hence

$$\begin{aligned} |\langle f, g \rangle| &= \left| \sum_j \langle P_j f, \tilde{P}_j g \rangle \right| \leq \int \sum_j |P_j f \cdot \tilde{P}_j g| \leq \int \left(\sum_j |P_j f|^2 \right)^{1/2} \left(\sum_j |\tilde{P}_j g|^2 \right)^{1/2} \\ &= \int S_P f \cdot \tilde{S}_P g \leq \|S_P f\|_{L^p} \|\tilde{S}_P g\|_{L^{p'}} \leq C_{d,p} \|S_P f\|_p \|g\|_{p'}. \end{aligned}$$

Taking $\|g\|_{p'} = 1$ and the supremum over such g ,

$$\|f\|_p \leq C_{p,d} \|S_P f\|_p. \quad \blacksquare$$

17 Applications of Littlewood–Paley Theory

17.1 Homogeneous Sobolev spaces and Sobolev embedding

DEF. For $s \geq 0$, the **homogeneous Sobolev space** $\dot{H}^s(\mathbb{R}^d)$ is the completion of $\mathcal{S}(\mathbb{R}^d)$ under

$$\|f\|_{\dot{H}^s(\mathbb{R}^d)} = \left(\int |\xi|^{2s} |\widehat{f}(\xi)|^2 d\xi \right)^{1/2}.$$

PROPOSITION. Let $0 \leq s < \frac{d}{2}$ and $\frac{1}{2} - \frac{1}{p} = \frac{s}{d}$. Then

$$\|f\|_{L^p(\mathbb{R}^d)} \leq C_{p,d} \|f\|_{\dot{H}^s} \quad \forall f \in \mathcal{S}(\mathbb{R}^d).$$

Proof.

SPECIAL CASE: $f \in \mathcal{S}(\mathbb{R}^d)$ with $\text{supp } \widehat{f} \subseteq \{\frac{R}{4} \leq |\xi| \leq 4R\}$. For such f ,

$$\|f\|_{\dot{H}^s}^2 = \int |\xi|^{2s} |\widehat{f}(\xi)|^2 \geq C_s R^{2s} \int |\widehat{f}|^2 = C_s R^{2s} \int |f|^2.$$

So it suffices to show

$$\|f\|_{L^p(\mathbb{R}^d)} \leq C_{p,d} R^s \|f\|_{L^2(\mathbb{R}^d)}, \quad s = d\left(\frac{1}{2} - \frac{1}{p}\right). \quad (**)$$

Define $g(x) = f(Rx) R^d \Rightarrow \widehat{g}(\xi) = \widehat{f}(\xi/R)$. Then $(**)$ is equivalent to

$$\|g\|_{L^p} \leq C_{p,d} \|g\|_{L^2}.$$

Find $\chi \equiv 1$ on $\frac{1}{4} \leq |x| \leq 4$; then $\widehat{g}(\xi) \cdot \chi = \widehat{g}$, so $g = \check{\chi} * g$. By **Young's inequality** (with $1 + \frac{1}{p} = \frac{1}{2} + \frac{1}{q}$),

$$\|g\|_{L^p} = \|\check{\chi} * g\|_p \leq \|\check{\chi}\|_q \|g\|_2.$$

GENERAL CASE:

$$\begin{aligned} \|f\|_p &\approx \|S_P f\|_p = \left\| \left(\sum_j |P_j f|^2 \right)^{1/2} \right\|_p \leq \left(\sum_j \|P_j f\|_p^2 \right)^{1/2} \\ &\lesssim \left(\sum_j \underbrace{2^{2js} \|P_j f\|_{L^2}^2} \right)^{1/2} \leq \left(\sum_j \int |\widehat{P_j f}|^2 |\xi|^{2s} \right)^{1/2} = \|f\|_{\dot{H}^s}. \quad \blacksquare \end{aligned}$$

(Middle step: apply the special case $(**)$ to $P_j f$, which has frequency $\sim 2^j$, giving $\|P_j f\|_p \lesssim 2^{js} \|P_j f\|_2$.)

17.2 Hölder spaces and their Littlewood–Paley characterization

HÖLDER SPACE.

$$C^\alpha(\mathbb{R}^n) = \left\{ f \mid [f]_\alpha = \sup_{x,y \in \mathbb{R}^d} \frac{|f(x+y) - f(x)|}{|y|^\alpha} < +\infty \right\}.$$

THEOREM. Suppose $|f(x)| \leq 1$. Then

$$f \in C^\alpha \iff \exists A \text{ s.t. } \|P_j f\|_{L^\infty(\mathbb{R}^d)} \leq A \cdot 2^{-j\alpha},$$

and moreover

$$C_{d,\alpha}^{-1} A \leq [f]_\alpha \leq C_{d,\alpha} A.$$

17.2.1 Proof

($\Rightarrow$ DIRECTION / KERNEL SIDE.) Since $P_j f = \check{\psi}_j * f$ and ψ_j is supported away from 0:

$$D^r \psi_j \in C_c^\infty(\mathbb{R}^d \setminus \{0\}) \Rightarrow D^r \psi_j(0) \equiv 0 \quad \forall r \Rightarrow \int x^r \check{\psi}_j(x) dx \equiv 0 \quad \forall r.$$

In particular $\int \check{\psi}_j = 0$, so

$$P_j f(x) = \check{\psi}_j * f = \int \check{\psi}_j(y) f(x-y) dy = \int \check{\psi}_j(y) (f(x-y) - f(x)) dy.$$

Hence, using $\check{\psi}_j(y) = 2^{jd} \check{\psi}(2^j y)$ and the change of variables $u = 2^j y$,

$$\begin{aligned} |P_j f(x)| &\leq \int |\check{\psi}_j(y)| |f(x-y) - f(x)| dy \\ &\leq [f]_\alpha \int |\check{\psi}_j(y)| |y|^\alpha dy \\ &= [f]_\alpha \int 2^{jd} |\check{\psi}(2^j y)| |y|^\alpha dy \\ &= C 2^{-j\alpha} [f]_\alpha. \end{aligned}$$

($\Leftarrow$ DIRECTION / CONVERSE.)

LEMMA (BERNSTEIN-TYPE). Define $Pf(x) = g * f$, where $f \in L^\infty$ and g satisfies $\text{supp } \hat{g} \subset \{|\xi| < R\}$. Then

$$\|\nabla Pf\|_{L^p} \leq R \cdot C_d \|Pf\|_{L^p}, \quad 1 \leq p \leq \infty.$$

Proof. Since $\text{supp}(\hat{g}) \subseteq \{|\xi| \leq R\}$, we have $\hat{g}(\xi) = \hat{g}(\xi) \cdot \chi(\xi/R)$, i.e. $g = g * \check{\chi}_R$, so $\nabla g = g * \nabla \check{\chi}_R$. With

$$\check{\chi}_R(x) = R^d \check{\chi}(Rx), \quad \nabla \check{\chi}_R(x) = R^{d+1} (\nabla \check{\chi})(Rx),$$

we get $\nabla Pf = \nabla(g * f) = \nabla \check{\chi}_R * g * f$, hence

$$\|\nabla \check{\chi}_R * g * f\|_{L^p} \leq \|\nabla \check{\chi}_R\|_{L^1} \|Pf\|_{L^p} = R \|\nabla \check{\chi}\|_{L^1} \|Pf\|_{L^p}. \quad \square$$

Now let f satisfy $\|P_j f\|_{L^\infty} \leq A 2^{-j\alpha}$, and set $g_\ell = \sum_{-j \leq \ell} P_j f$. Decompose the increment into a low-frequency gradient term and a high-frequency tail:

$$|g_\ell(x+y) - g_\ell(x)| \leq \underbrace{\sum_{|j| \leq \ell} |P_j f(x+y) - P_j f(x)|}_{L_\ell(x,y)} + \underbrace{\sum_{j > \ell} 2 \|P_j f\|_\infty}_{H_\ell}.$$

HIGH-FREQUENCY ESTIMATE.

$$H_\ell \leq 2A \sum_{j \geq \ell} 2^{-j\alpha} \lesssim A 2^{-\ell\alpha}.$$

LOW-FREQUENCY ESTIMATE.

$$\begin{aligned} |P_j f(x+y) - P_j f(x)| &= \left| \int_0^1 (\nabla P_j f) \cdot y \, dt \right| \\ &\leq \|\nabla P_j f\|_\infty |y| \leq 2^j \|P_j f\|_\infty |y| \\ &\leq A 2^{j(1-\alpha)} |y|. \end{aligned}$$

Summing,

$$L_\ell(x, y) \leq \sum_{|j| \leq \ell} A 2^{j(1-\alpha)} |y| \lesssim A 2^{\ell(1-\alpha)} |y|.$$

OPTIMIZE. Take $2^\ell = \frac{1}{|y|}$. Then

$$H_\ell \lesssim A |y|^\alpha, \quad L_\ell(x, y) \lesssim A |y|^\alpha.$$

So $|g_\ell(x+y) - g_\ell(x)| \leq CA |y|^\alpha$, i.e. $[g_\ell]_\alpha \leq CA$.

Since $g_\ell \rightarrow f$, we conclude $[f]_\alpha \leq CA$. ■

Part VI

Oscillatory Integrals and Fourier Restriction

18 Oscillatory Integrals and Stationary Phase

18.1 Fourier transforms of surface measures

Consider the **Schrödinger equation**

$$\begin{cases} i\partial_t u + \Delta u = 0, \\ u(x, 0) = u_0(x), \end{cases} \quad u_0 \in \mathcal{S}(\mathbb{R}^d).$$

On the Fourier side:

$$\begin{cases} i\partial_t \hat{u} - 4\pi^2 |\xi|^2 \hat{u} = 0, \\ \hat{u}(\xi, 0) = \hat{u}_0(\xi), \end{cases} \quad \implies \quad \hat{u}(\xi, t) = e^{-4\pi^2 i |\xi|^2 t} \hat{u}_0(\xi).$$

Hence

$$\begin{aligned} u &= \mathcal{F}_\xi^{-1}(e^{-4\pi^2 i |\xi|^2 t} \hat{u}_0(\xi))(x) = \int e^{2\pi i x \cdot \xi} e^{-4\pi^2 i |\xi|^2 t} \hat{u}_0(\xi) \, d\xi \\ &= \int_{\mathbb{R}^d} e^{2\pi i (x, t) \cdot (\xi, -2\pi |\xi|^2)} \hat{u}_0(\xi) \, d\xi = \int_{\mathbb{R}^{d+1}} e^{2\pi i (x, t) \cdot (\xi, \tau)} \hat{u}_0(\xi) \, d\mu(\xi, \tau), \end{aligned}$$

where μ is the measure on the paraboloid defined (as a distribution) by

$$\int F(\xi, \tau) \, d\mu(\xi, \tau) = \int_{\mathbb{R}^d} F(\xi, -2\pi |\xi|^2) \, d\xi.$$

Thus μ is supported on the downward paraboloid $\tau = -2\pi|\xi|^2$.

This representation raises the question of how rapidly the Fourier transform of the surface measure decays. At positive times the initially concentrated profile spreads and oscillates.

One can show $u_0 \in \mathcal{S}(\mathbb{R}^d) \Rightarrow u(\cdot, t) \in \mathcal{S}(\mathbb{R}^d)$ for $t > 0$:

$$u(x, t) = C_d \frac{1}{t^{d/2}} \int e^{i|x-y|^2/4t} u_0(y) dy \implies \|u(\cdot, t)\|_\infty \leq C_d t^{-d/2} \|u_0\|_{L^1}.$$

$u(\cdot, t)$ decays faster than $|x|^{-N}$ for all N , because it is Schwartz in x . Therefore

$$|\hat{\mu}(x, t)| \leq C_d \|(x, t)\|^{-d/2}, \quad d = \dim(\text{hypersurface}).$$

CONJECTURE: $|\hat{\mu}(x)| \lesssim |x|^{-d/2}$.

The conjectured decay fails for flat hypersurfaces. For example, in $\mathbb{R}^{d+1}$ take $\mu(\xi, \tau) = d\xi \times \delta_0(\tau)$. Then

$$\mathcal{F}^{-1}(u d\mu) = \int e^{2\pi i(x, t) \cdot (\xi, \tau)} u(\xi) d\xi \delta_0 d\tau = \int e^{2\pi i x \cdot \xi} u(\xi) d\xi$$

has no decay in the t direction, precisely because there is no curvature in that direction. Non-vanishing Gaussian curvature restores decay:

Theorem 18.1 (Decay of curved surface measure). *Let $M \subseteq \mathbb{R}^d$ be a hypersurface with non-vanishing Gauss curvature, and let $d\mu = \psi d\sigma$, where $\psi \in C_c^\infty$ and $d\sigma$ is surface measure. Then*

$$|\hat{\mu}(\xi)| \leq C(M, \psi) |\xi|^{-\frac{d-1}{2}}.$$

18.2 Stationary phase

The proof rests on integrals of the form

$$I(\lambda) = \int e^{i\lambda\phi(\xi)} a(\xi) d\xi, \quad a \in C_c^\infty.$$

There are two distinct regimes:

1. If $\nabla\phi \neq 0$ on $\text{supp}(a)$, repeated integration by parts gives rapid decay.
2. If $\nabla\phi(\xi_0) = 0$ for some ξ_0 , the main contribution comes from a neighbourhood of the critical point.

18.2.1 Non-stationary phase

If $\nabla\phi \neq 0$ for all $\xi \in \text{supp}(a)$, then

$$|I(\lambda)| = \left| \int e^{i\lambda\phi} a(\xi) d\xi \right| \leq C_{\phi, a, N} \lambda^{-N} \quad \forall N.$$

PROOF (INTEGRATION BY PARTS).

Case $d = 1$. Since $[e^{i\lambda\phi}]' = i\lambda\phi' e^{i\lambda\phi}$,

$$\int e^{i\lambda\phi} a = \int \frac{1}{i\lambda\phi'} [e^{i\lambda\phi}]' a = \frac{1}{i\lambda} \int [e^{i\lambda\phi}]' \frac{a}{\phi'} = -\frac{1}{i\lambda} \int e^{i\lambda\phi} \left[\frac{a}{\phi'} \right]' = \frac{1}{i\lambda} \int e^{i\lambda\phi} a_1,$$

and repeat several times (each pass gains a factor λ^{-1}).

Case $d \geq 2$.

$$\nabla e^{i\lambda\phi} = e^{i\lambda\phi} \nabla\phi \cdot i\lambda \implies \nabla\phi \cdot \nabla e^{i\lambda\phi} = e^{i\lambda\phi} |\nabla\phi|^2 i\lambda \implies e^{i\lambda\phi} = \frac{\nabla\phi \cdot \nabla e^{i\lambda\phi}}{|\nabla\phi|^2} \cdot \frac{1}{i\lambda},$$

$$\implies \int e^{i\lambda\phi} a \, d\xi = \int \frac{\nabla\phi \cdot \nabla e^{i\lambda\phi}}{|\nabla\phi|^2} \frac{a}{i\lambda} = \frac{1}{i\lambda} \left(- \int \operatorname{div} \left(a \frac{\nabla\phi}{|\nabla\phi|^2} \right) e^{i\lambda\phi} \, d\xi \right).$$

18.2.2 The stationary regime

For a stationary point we need **degeneracy** of ϕ at ξ_0 .

Example: $\phi \equiv 0 \Rightarrow I(\lambda) = \int a(\xi) \, d\xi$ — no decay.

We need $\det(D^2\phi(\xi_0)) \neq 0$.

Theorem 18.2 (Stationary phase). *Suppose $\nabla\phi(\xi_0) = 0$, $\nabla\phi \neq 0$ on $\operatorname{supp}(a) \setminus \{\xi_0\}$, and $\det D^2\phi(\xi_0) \neq 0$. Then*

$$(1) \quad |I(\lambda)| = \left| \int_{\mathbb{R}^d} e^{i\lambda\phi} a(\xi) \, d\xi \right| \leq C \lambda^{-d/2},$$

$$(2) \quad |\partial_\lambda^k (e^{-i\lambda\phi(\xi_0)} I(\lambda))| \leq C \lambda^{-\frac{d}{2}-k}.$$

PROOF. Assume $\xi_0 = 0$. Write (with $\chi = 1$ on $\{|\xi| \leq 1\}$, $\chi = 0$ on $\{|\xi| \geq 2\}$)

$$\int_{\mathbb{R}^d} e^{i\lambda\phi(\xi)} a(\xi) \, d\xi = \underbrace{\int e^{i\lambda\phi} \chi(\xi/\varepsilon) a \, d\xi}_A + \underbrace{\int (1 - \chi(\xi/\varepsilon)) e^{i\lambda\phi} a \, d\xi}_B.$$

18.2.3 Estimating the main and error terms

ONE-DIMENSIONAL MODEL.

$$A(\lambda) \leq \int |\chi(\xi/\varepsilon)| a(\xi) \, d\xi \leq \varepsilon \cdot C,$$

$$B(\lambda) \stackrel{\text{IBP}}{=} \left(\frac{1}{i\lambda} \right)^N \int e^{i\lambda\phi(\xi)} L^N((1 - \chi(\xi/\varepsilon))a(\xi)) \, d\xi, \quad Lf(\xi) = \left(\frac{f(\xi)}{\phi'(\xi)} \right)'.$$

Applying L^N produces three types of terms:

1. derivatives landing on a ;
2. derivatives landing on $(1 - \chi) \rightsquigarrow$ factors of ε^{-N} ;
3. factors from $1/\phi' \rightsquigarrow \frac{1}{|\xi|^{2N}}$ (since $\phi'(\xi) \sim |\xi|$ near 0).

Hence

$$|B(\lambda)| \leq C \int_{|\xi| \geq \varepsilon} \left(\frac{1}{|\xi|^{2N}} + \frac{\varepsilon^{-N}}{|\xi|^N} \right) d\xi \approx \varepsilon^{-(2N-1)},$$

so $|B| \lesssim \lambda^{-N} \varepsilon^{-(2N-1)}$. Balancing with $|A| \lesssim \varepsilon$ via $\varepsilon^{2N} = \lambda^{-N}$ gives

$$\boxed{\varepsilon = \lambda^{-1/2}}.$$

HIGHER DIMENSION, $d \geq 2$, $\xi_0 = 0$, $\det(D^2\phi) \neq 0 \Rightarrow |\nabla\phi(\xi)| \sim |\xi|$ (use Taylor expansion). Let χ be a smooth bump, $\chi(\xi) = 1$ if $|\xi| \leq 1$, $\chi(\xi) = 0$ if $|\xi| \geq 2$:

$$I(\lambda) = \underbrace{\int e^{i\lambda\phi} \chi(\xi/\varepsilon) a \, d\xi}_{A(\lambda)} + \underbrace{\int e^{i\lambda\phi} (1 - \chi(\xi/\varepsilon)) a \, d\xi}_{B(\lambda)}.$$

$$|A(\lambda)| \leq \int |\chi(\xi/\varepsilon) a(\xi)| \, d\xi \leq C \int \chi(\xi/\varepsilon) \, d\xi \leq C\varepsilon^d.$$

For $B(\lambda)$, integrate by parts:

$$B(\lambda) = \int e^{i\lambda\phi(\xi)} (1 - \chi(\xi/\varepsilon)) a = \left(\frac{-1}{i\lambda}\right)^N \int e^{i\lambda\phi} L^N((1 - \chi(\xi/\varepsilon)) a) \, d\xi, \quad Lf = \operatorname{div}\left(\frac{\nabla\phi f(\xi)}{|\nabla\phi|^2}\right).$$

Lemma 18.3. *On $\{|\xi| \geq \varepsilon\}$ one has*

$$\left| L^N((1 - \chi(\xi/\varepsilon)) a(\xi)) \right| \leq C \frac{\varepsilon^{-N}}{|\xi|^N}.$$

Given the claim,

$$|B(\lambda)| \leq \frac{1}{\lambda^N} \int_{|\xi| \geq \varepsilon} \varepsilon^{-N} \frac{1}{|\xi|^N} \, d\xi \leq \frac{1}{\lambda^N} \varepsilon^{-N} \int_{\geq \varepsilon} r^{d-1} r^{-N} \, dr = \frac{\varepsilon^{-N}}{\lambda^N} \varepsilon^{-N+d} = \frac{\varepsilon^{-2N+d}}{\lambda^N},$$

$$\Rightarrow |I| \leq \varepsilon^d + \frac{\varepsilon^{-2N+d}}{\lambda^N}; \quad \text{take } \varepsilon \text{ with } \varepsilon^{2N} = \lambda^{-N} \Rightarrow \varepsilon = \lambda^{-1/2}.$$

PROOF OF CLAIM.

$$L^N((1 - \chi(\xi/\varepsilon)) a) = \cdots \left(\operatorname{div} \frac{\nabla\phi}{|\nabla\phi|^2} \operatorname{div} \left(\frac{\nabla\phi(1-\chi)}{|\nabla\phi|^2} a \right) \right).$$

- *Case 1:* derivative on a .
- *Case 2:* derivative on $\chi(\xi/\varepsilon) \rightsquigarrow \varepsilon^{-N}$.
- *Case 3:* on $\frac{\nabla\phi}{|\nabla\phi|^2}$:

$$\partial_\ell \left(\frac{\nabla\phi}{|\nabla\phi|^2} \right) = \frac{\nabla\partial_\ell\phi}{|\nabla\phi|^2} - \nabla\phi \cdot \frac{2\langle \nabla\partial_\ell\phi, \nabla\phi \rangle}{|\nabla\phi|^4} \sim \frac{1}{|\nabla\phi|^2} \rightsquigarrow \frac{1}{|\nabla\phi|^{2N}}.$$

18.2.4 Completion of the estimate

(Note: each step adds one more factor; the two parts together account for the N derivatives.)

$$|I| \leq C \sum_{k=0}^N \varepsilon^{-k} \left(\sum_{\ell=0}^{N-k} \frac{1}{|\nabla\phi|^{N+\ell}} \right).$$

If $|\xi| \geq \varepsilon$ (so $k + \ell \leq N$),

$$\leq C \sum_{k=0}^N \frac{1}{|\xi|^N} \sum_{\ell=0}^{N-k} \varepsilon^{-(k+\ell)} \leq \frac{\varepsilon^{-N}}{|\xi|^N} \cdot C, \quad \varepsilon^{-(k+\ell)} = \varepsilon^{-N} \cdot \underbrace{\varepsilon^{N-(k+\ell)}}_{\leq 1}.$$

PART 2 (DERIVATIVE BOUND).

$$I(\lambda) = \int e^{i\lambda\phi} a(\xi) d\xi = e^{i\lambda\phi(\xi_0)} \int_{\mathbb{R}^d} e^{i\lambda(\phi-\phi(\xi_0))} a(\xi) d\xi$$

$$\Rightarrow e^{-i\lambda\phi(\xi_0)} I(\lambda) = \int e^{i\lambda(\phi-\phi(\xi_0))} a(\xi) d\xi$$

$$\Rightarrow \frac{d}{d\lambda}(\dots) = \int e^{i\lambda(\phi-\phi(\xi_0))} (\phi-\phi(\xi_0)) a(\xi) \cdot i d\xi = \frac{-1}{i\lambda} \int e^{i\lambda(\phi-\phi(\xi_0))} \operatorname{div} \left(\frac{\nabla \phi}{|\nabla \phi|^2} i(\phi-\phi(\xi_0)) a \right) d\xi,$$

where $\phi - \phi(\xi_0) \approx |\xi|^2$ and the integrand is C_c^∞ . Each λ -derivative gains λ^{-1} .

18.3 Fourier decay of surface measures

In $\mathbb{R}^{d+1}$ let M be a smooth hypersurface, $d\mu = \psi d\sigma$ with $\psi \in C_c^\infty(\mathbb{R}^{d+1})$, $d\sigma$ the surface measure, and $N(x)$ = normal vector at x . For $\xi \in \mathbb{R}^{d+1}$:

1. If ξ is **not parallel** to $N(x)$ for all $x \in \operatorname{supp} \psi \cap M$, then $|\hat{\mu}(\xi)| \leq C_{\psi, M} |\xi|^{-k} \forall k \geq 0$.
2. If M has **non-vanishing curvature**, then $|\hat{\mu}(\xi)| \leq C_{\psi, M} |\xi|^{-d/2}$.

PROOF (SETUP). Cover $M \cap \operatorname{supp} \psi$ by balls (B_i) such that on $\frac{1}{2}B_i$, M is a graph $f_1, \dots, f_n$. Take a finite subcover and a partition of unity $\{\psi_i\}$ subordinate to (B_i) with $\psi_i = 1$ on $\frac{1}{2}B_i$, $\operatorname{supp} \psi_i \subset B_i$. Then

$$\hat{\mu}(\xi) = \int e^{-2\pi i x \cdot \xi} d\mu(x) = \sum_i \int e^{-2\pi i x \cdot \xi} \underbrace{\psi_i \psi(x) d\sigma(x)}_{d\mu_i(x)}.$$

It suffices to show $|\hat{\mu}_i(\xi)| \leq C_{\psi, M} |\xi|^{-d/2}$. Writing M as a graph $x = (y, f(y))$,

$$\hat{\mu}_i(\xi) = \int e^{-2\pi i x \cdot \xi} \psi_i \psi d\sigma = \int_{\mathbb{R}^d} e^{-2\pi i (y, f(y)) \cdot \xi} \underbrace{\psi_i(y, f(y)) \psi(y, f(y)) \sqrt{1 + |\nabla f|^2}}_{a(y)} dy,$$

which "looks like the integral before."

19 Fourier Restriction Theory

19.1 Critical points and the restriction problem

Write $\xi = \lambda\nu$:

$$\hat{\mu}_i(\xi) = \int_{\mathbb{R}^d} e^{-2\pi i \lambda (y, f(y)) \cdot \nu} \psi_i(y, f(y)) \psi(y, f(y)) \sqrt{1 + |\nabla f|^2} dy.$$

1. WHEN DOES THE PHASE HAVE A CRITICAL POINT? Consider $x \mapsto -\langle x, \xi \rangle$ restricted to M . It has a critical point at $x_0 \in M$

$$\iff \forall \text{ curves } r : (-1, 1) \rightarrow M, r(0) = x_0, t \mapsto -\langle r(t), \xi \rangle \text{ is critical at } 0$$

$$\Longleftrightarrow -\langle r'(0), \xi \rangle = 0 \Longleftrightarrow \xi \parallel N(x_0).$$

So if there is **no** critical point $\Rightarrow$ non-stationary phase (case 1 above).

2. THERE IS A CRITICAL POINT AT $x_0 \Longleftrightarrow \xi \parallel N(x_0)$.

- Only finitely many such points can occur.
- Refine the balls so that at most one ball meets each critical point:

$$\hat{\mu}_i = \int_{\mathbb{R}^n} e^{-2\pi i \lambda (y, f(y)) \cdot \nu} \underbrace{\psi_i \psi}_a dy, \quad \xi \parallel N(x_0).$$

With $\nu = N(x_0) = (-\partial_1 f, \dots, -\partial_d f, 1) / \sqrt{1 + |\nabla f(x_0)|^2}$, the phase is

$$\phi(y) = (y, f(y)) \cdot (-\partial_1 f(y_0), \dots, -\partial_d f(y_0), 1),$$

$$\partial_i \partial_j \phi = (0, \partial_{ij} f) \cdot (\dots, 1) / \sqrt{1 + |\nabla f|^2} \implies \det(\phi_{ij}) \neq 0 \Longleftrightarrow \det(\partial_{ij} f) \neq 0.$$

So non-degenerate Hessian of f (non-vanishing Gaussian curvature) $\Longleftrightarrow$ non-degenerate stationary point, giving the $\lambda^{-d/2}$ decay.

19.1.1 Formulation of the restriction problem

For $f \in L^1(\mathbb{R}^d)$, the Fourier transform $\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x \cdot \xi} dx$ is continuous and decays at infinity.

- $L^1(\mathbb{R}^d) \xrightarrow{\mathcal{F}} C_0(\mathbb{R}^d)$, with $\|\hat{f}\|_{L^\infty} \leq \|f\|_{L^1}$.
- If $f \in L^1 \cap L^2$, then $\|\hat{f}\|_{L^2} = \|f\|_{L^2}$ (Plancherel).
- By Riesz–Thorin: $\|\hat{f}\|_{p'} \leq \|f\|_p$ for $1 \leq p \leq 2$.

THE RESTRICTION PROBLEM asks, for given $1 \leq p \leq 2$ and $S \subseteq \mathbb{R}^d$, whether $\hat{f}|_S$ is meaningful.

- If S has positive measure, $\hat{f} \in L^{p'} \Rightarrow$ meaningful.
- If S has measure 0: for $p = 1$, $\hat{f} \in C_0(\mathbb{R}^d)$; but for $p = 2$, $\hat{f} \in L^2(\mathbb{R}^d)$ only, so restriction is **impossible**.

19.2 The L^2 restriction theorem

STEIN'S OBSERVATION. If $S = S^{d-1} \subseteq \mathbb{R}^d$, then restriction holds for $f \in L^p$, $1 < p < 2$: i.e. $\hat{f}$ has more structure in $L^{p'}$ than the trivial bound suggests.

Theorem 19.1 (L^2 restriction). *Let $d \geq 2$ and $1 \leq p < \frac{4d}{3d+1}$. Then*

$$\|\hat{f}\|_{L^2(S^{d-1}, d\sigma)} \lesssim_{d,p} \|f\|_{L^p(\mathbb{R}^d)} \quad \forall f \in \mathcal{S}(\mathbb{R}^d),$$

where $d\sigma$ is surface measure.

PROOF. Fix d, p, f :

$$\begin{aligned}\|\widehat{f}\|_{L^2(S^{d-1}, d\sigma)}^2 &= \int_{S^{d-1}} |\widehat{f}|^2 d\sigma = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \int_{S^{d-1}} f(x) \overline{f(y)} e^{-2\pi i(x-y)\cdot\xi} dx dy d\sigma(\xi) \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x) \overline{f(y)} (d\sigma)^\vee(y-x) dx dy = \langle f, f * (d\sigma)^\vee \rangle.\end{aligned}$$

Now $(d\sigma)^\vee(x) \lesssim 1$ trivially. By rotation invariance, with $x = (0, \dots, |x|)$,

$$I(x) = (d\sigma)^\vee(x) = \int_{\mathbb{R}^d} e^{2\pi i \xi \cdot x} d\sigma(\xi) = \int_{\mathbb{R}^d} e^{2\pi i \xi_d \cdot |x|} d\sigma(\xi)$$

(the height function is ξ_d). By the (non-)stationary phase method, splitting the sphere into three parts — *good parts* ($\lesssim |x|^{-N}$) and *bad parts* (the two poles) — one gets

$$(d\sigma)^\vee(x) \lesssim_d |x|^{-\frac{d-1}{2}}, \quad |x| \geq 1, \quad \text{i.e.} \quad (d\sigma)^\vee(x) \lesssim_d \langle x \rangle^{-\frac{d-1}{2}}.$$

Since $\langle x \rangle^{-a} \in L^p \iff p > \frac{d}{a}$, we get $(d\sigma)^\vee \in L^q$ for $q > \frac{2d}{d-1}$.

By Young's inequality ($\|f * g\|_{L^r} \leq \|f\|_{L^p} \|g\|_{L^q}$, $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$),

$$\|\widehat{f}\|_{L^2(S^{d-1}, d\sigma)}^2 = \langle f, f * (d\sigma)^\vee \rangle, \quad \|f * (d\sigma)^\vee\|_{L^{p'}} \lesssim \|f\|_{L^p},$$

$$|\langle f, f * (d\sigma)^\vee \rangle| \leq \|f\|_{L^p(\mathbb{R}^d)} \|f * (d\sigma)^\vee\|_{L^{p'}(\mathbb{R}^d)} \lesssim \|f\|_{L^p}^2. \quad \blacksquare$$

Conjecture 19.2 (Stein). *For $d \geq 2$,*

$$\|\widehat{f}\|_{L^q(S^{d-1}, d\sigma)} \lesssim_{d,p,f} \|f\|_{L^p}, \quad p < \frac{2d}{d+1}, \quad q \leq \frac{d-1}{d+1} p'.$$

19.3 Restriction–extension duality

$d = 2$: SOLVED.

19.3.1 The dual formulation

Let μ be a Radon measure on $\mathbb{R}^d$, $1 \leq p, q \leq \infty$, $A > 0$. The following are equivalent.

- **(1) Restriction estimate** ($f \in C_c^\infty(\mathbb{R}^d)$):

$$\|\widehat{f}\|_{L^q(\mathbb{R}^d, \mu)} \leq A \|f\|_{L^p}.$$

- **(2) Extension estimate** ($g \in C_c^\infty$):

$$\|(g d\mu)^\vee\|_{L^{p'}} \leq A \|g\|_{L^{q'}(\mathbb{R}^d, \mu)}.$$

PROOF. (1) $\Rightarrow$ (2).

$$f \mapsto \int f \cdot \overline{(g d\mu)^\vee} dx = \int \widehat{f} g d\mu \leq \|\widehat{f}\|_{L^q(\mathbb{R}^d, \mu)} \|g\|_{L^{q'}(d\mu)} \leq A \|f\|_{L^p} \|g\|_{L^{q'}(d\mu)}$$

$$\Rightarrow (g d\mu)^\vee \in L^{p'} \text{ bounded by } A \|g\|_{L^{q'}(d\mu)}.$$

(2) $\Rightarrow$ (1).

$$g \mapsto \int f \overline{(g d\mu)^\vee} = \int \hat{f} g d\mu,$$

and

$$\begin{aligned} \|(g d\mu)^\vee\|_{L^{p'}} \|f\|_{L^p} &\leq A \|g\|_{L^{q'}(\mathbb{R}^d, \mu)} \|f\|_{L^p} \\ &\Rightarrow \|\hat{f}\|_{L^q} \leq A \|f\|_{L^p}. \end{aligned}$$

Write $R_{S^{d-1}}(p \rightarrow q)$ = "Stein's conjecture holds for p, q ". By duality,

$$\|(g d\sigma)^\vee\|_{L^{p'}} \leq A \|g\|_{L^{q'}(S^{d-1}, d\sigma)}, \quad \text{in particular } (g d\sigma)^\vee \in L^{p'}.$$

NECESSITY OF THE RANGE (adapting the bound for $(d\sigma)^\vee$ via the stationary-phase/SWM asymptotic):

$$\begin{aligned} (g d\sigma)^\vee(x) &= \frac{1}{|x|^{\frac{d-1}{2}}} \left(C_+ g\left(\frac{x}{|x|}\right) e^{2\pi i|x|} + C_- g\left(\frac{x}{|x|}\right) e^{-2\pi i|x|} \right) + O_{g,d}(|x|^{-\frac{d}{2}}), \\ &\Rightarrow \|(g d\sigma)^\vee\|_{L^{p'}}^{p'} \gtrsim \int_1^\infty r^{-\frac{d-1}{2}p'} r^{d-1} dr. \end{aligned}$$

For convergence we need $(-\frac{d-1}{2})p' + (d-1) < -1$, i.e.

$$p' > \frac{2d}{d-1} \Rightarrow \frac{1}{1-\frac{1}{p}} > \frac{2d}{d-1} \Rightarrow 1 - \frac{1}{p} < \frac{d-1}{2d} \Rightarrow \frac{1}{p} > \frac{d+1}{2d} \Rightarrow \boxed{p < \frac{2d}{d+1}}.$$

19.4 Necessary exponents, Stein's conjecture, and Kakeya

Proposition 19.3 (Preliminary L^2 restriction estimate). *Let $d \geq 2$ and $1 \leq p < \frac{4d}{3d+1}$. Then*

$$\|\hat{f}\|_{L^2(S^{d-1}, d\sigma)} \leq C_{p,d} \|f\|_{L^p(\mathbb{R}^d)} \quad \forall f \in \mathcal{S}(\mathbb{R}^d).$$

Conjecture 19.4 (Restriction to the sphere). *For $d \geq 2$,*

$$\|\hat{f}\|_{L^q(S^{d-1}, d\sigma)} \leq C_{d,p,q} \|f\|_{L^p(\mathbb{R}^d)} \quad \forall f \in C_c^\infty(\mathbb{R}^d),$$

$$\boxed{\forall p < \frac{2d}{d+1}, \quad q \leq \frac{d-1}{d+1} p'}.$$

Status: $d = 2$ is solved; for $d \geq 3$, only a partial range is known.

KAKEYA SET ($d \geq 2$): a set containing a unit line segment in every direction.

- **Conjecture:** Hausdorff dimension = d . (The notes here abbreviate; the standard Kakeya conjecture asserts full dimension d .)
- **Restriction $\Rightarrow$ Kakeya conjecture.**

19.4.1 Asymptotics of the extension operator

Assume $\|(g d\sigma)^\vee\|_{L^{p'}} \leq C\|g\|_{L^{q'}(S^{d-1}, d\sigma)}$. Then for all $g \in C_c^\infty(\mathbb{R}^d)$,

$$F(x) = (g d\sigma)^\vee(x) = \frac{1}{|x|^{\frac{d-1}{2}}} \left(C_+ g\left(\frac{x}{|x|}\right) e^{2\pi i|x|} + C_- g\left(\frac{x}{|x|}\right) e^{-2\pi i|x|} \right) + O(|x|^{-\frac{d}{2}}), \quad |x| \geq 1.$$

PROOF.

$$F(x) = \int_{S^{d-1}} g(\omega) e^{2\pi i x \cdot \omega} d\sigma(\omega).$$

Assume $x = (0, \dots, r)$ (using rotation invariance of $d\sigma$), so

$$F(x) = \int_{S^{d-1}} g(\omega) e^{2\pi i r \omega_d} d\omega.$$

Using a partition of unity localizing to the two poles and the rest:

$$F(x) = I_+(r) + I_-(r) + I_{\text{rest}}(r),$$

- (a) **Non-stationary (equator/rest):** $O(r^{-N})$.
- (b) **North pole:** parametrize $\omega(u) = (u, \sqrt{1 - |u|^2})$, $u \in \mathbb{R}^{d-1}$, $|u| \leq \delta$.

Then

$$d\sigma(u) = \sqrt{\det(D_u \omega(u)^\top D_u \omega(u))} du, \quad D_u \omega(u)^\top D_u \omega(u) = \text{Id} + O(|u|^2),$$

$$\omega_d = 1 - \frac{1}{2}|u|^2 + O(|u|^4) = \sqrt{1 - |u|^2},$$

$$a(u) = \chi_+(u) g(\omega(u)) (1 + O(|u|^2)), \quad \phi(u) = 2\pi \omega_d(u),$$

so $\phi(0) = 2\pi$, $\nabla \phi(0) = 0$, $D^2 \phi(0) = -2\pi \text{Id}$ (a non-degenerate maximum). Refined stationary phase (Lemma 1 below) then yields the stated asymptotic. ■

19.5 A refined stationary-phase lemma

Lemma 19.5 (Refined stationary phase).

$$\int e^{i\lambda \phi(u)} a(u) du = e^{i\lambda \phi(0)} e^{i\frac{\pi}{4} \text{sgn}(D^2 \phi(0))} \left| \det\left(\frac{1}{2\pi} D^2 \phi(0)\right) \right|^{-\frac{1}{2}} (a(0) + O(\lambda^{-1})).$$

Applying it at the north pole (with large parameter r , $\phi(0) = 1$, $D^2 \phi(0) = -2\pi \text{Id}$, $\text{sgn} = -(d-1)$):

$$I_+(r) = e^{2\pi i r} C_d r^{-\frac{d-1}{2}} (g(0, \dots, 1) + O(r^{-1})), \quad C_d = e^{-i\frac{\pi}{4}(d-1)} (2\pi)^{-\frac{d-1}{2}}.$$